

Design of Voltage Reference Circuit

Vinayak Hande

1

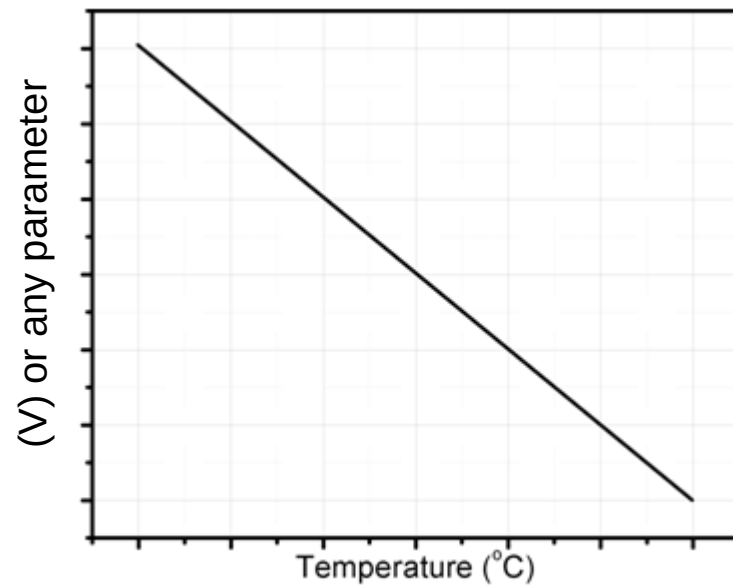


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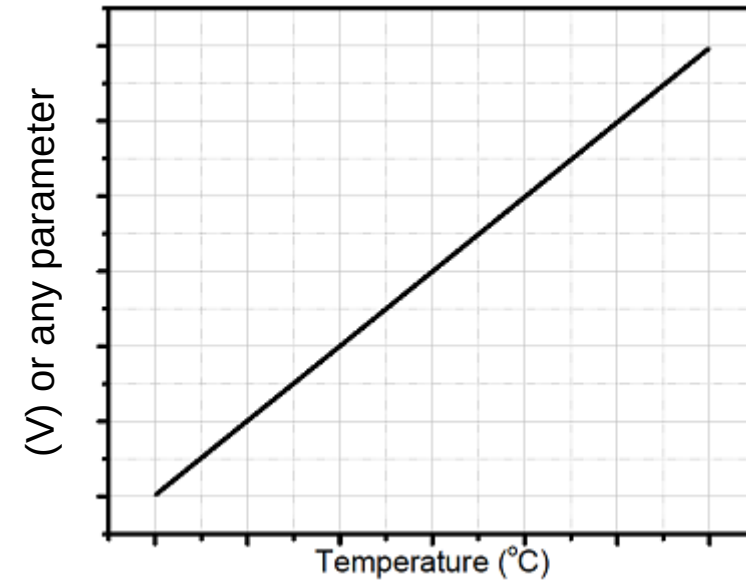
1. Recap
2. Optimisation
3. Error Sources
4. Higher order voltage references
5. Layout consideration
6. Conclusion : Part 2

Terminologies

(1) CTAT (Complementary to absolute temperature)

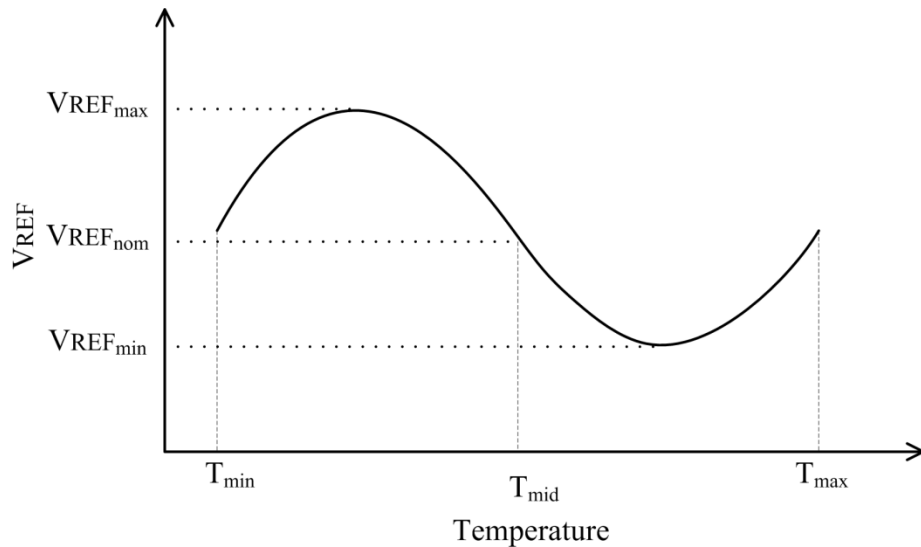


(2) PTAT (Proportional to absolute temperature)



Terminologies

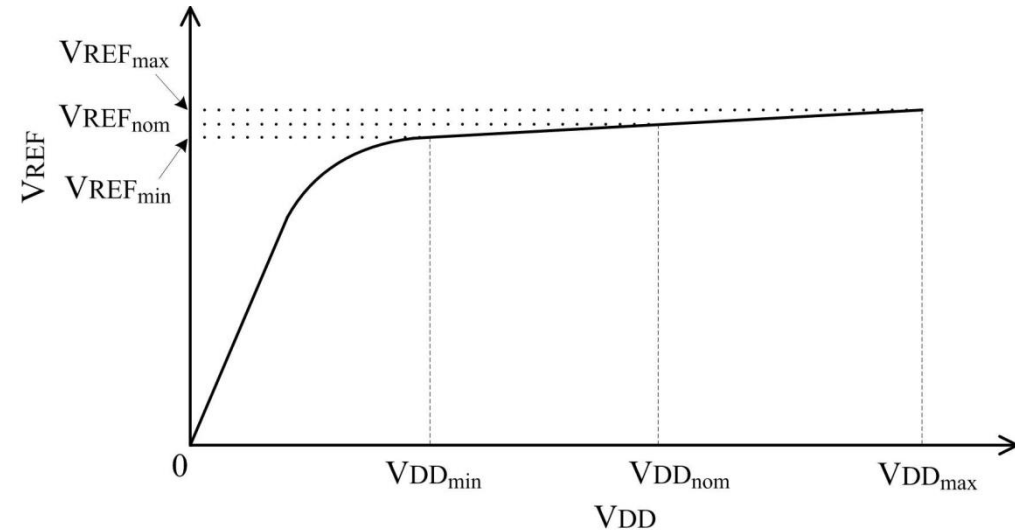
(3) Temperature coefficient (TC)



$$TC = \frac{V_{REF_{max}} - V_{REF_{min}}}{V_{REF_{nom}} (T_{max} - T_{min})} 10^6 \text{ (ppm/}^\circ\text{C)}$$

(Smaller the better)

(4) Line regulation (LR)

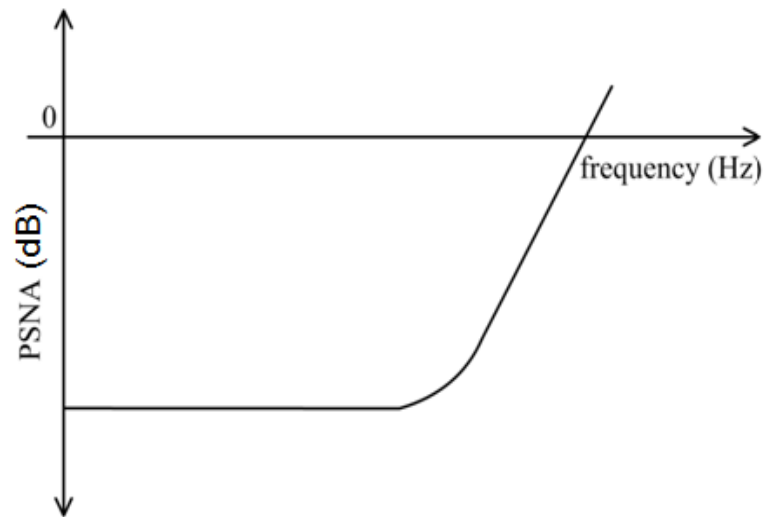


$$LR = \frac{V_{REF_{max}} - V_{REF_{min}}}{(V_{DD_{max}} - V_{DD_{min}})} \text{ (mV/V)}$$

(Smaller the better)

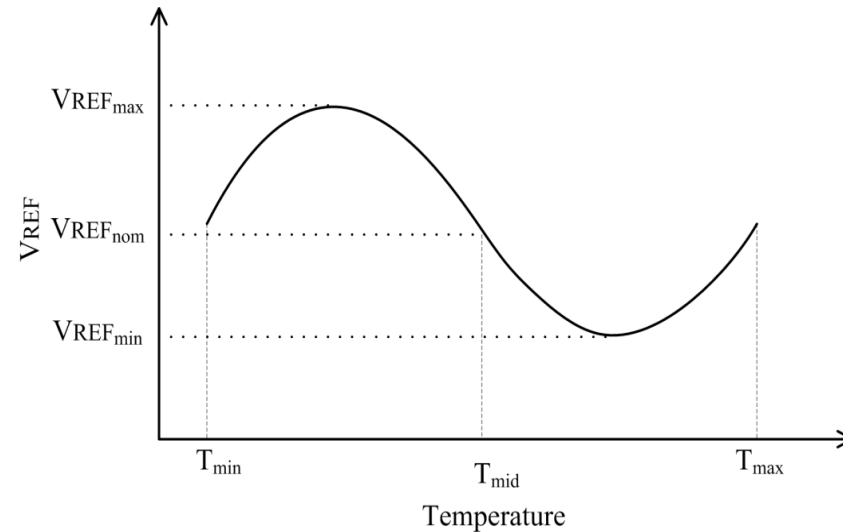
Terminologies

(5) Power supply noise attenuation (PSNA)



(Smaller the better)

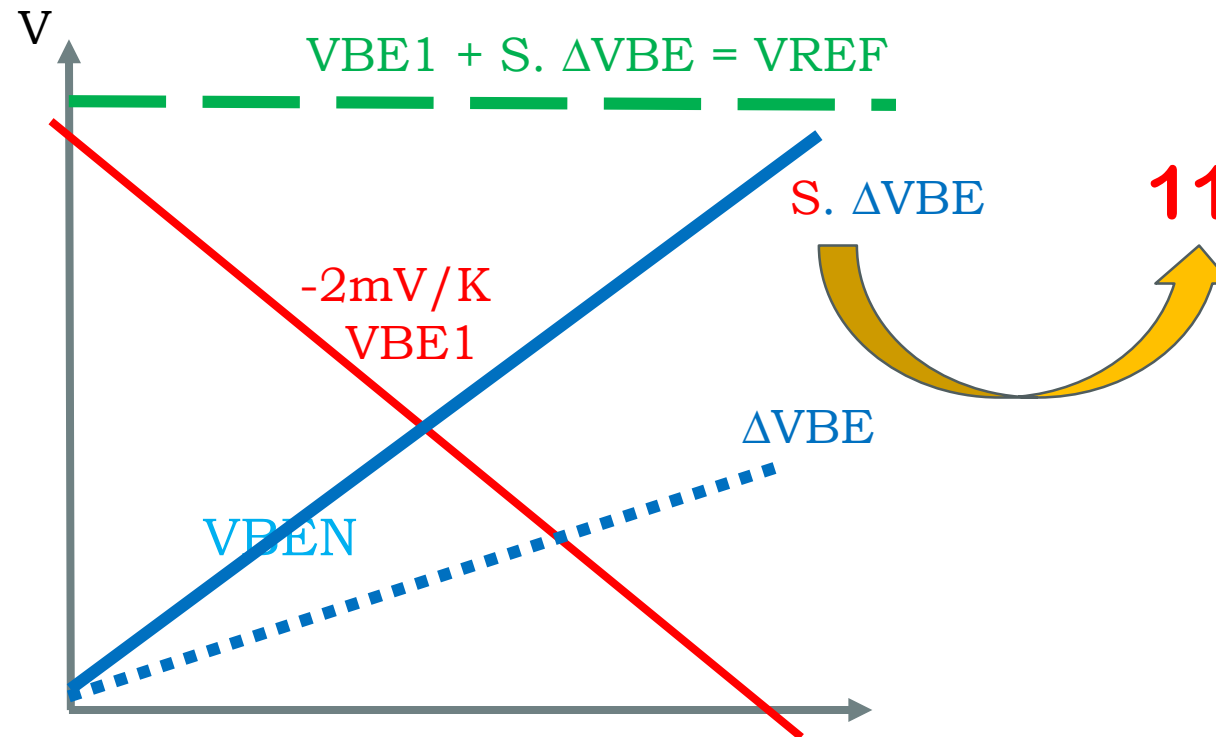
(6) Maximum absolute percentage variation (MAPV)



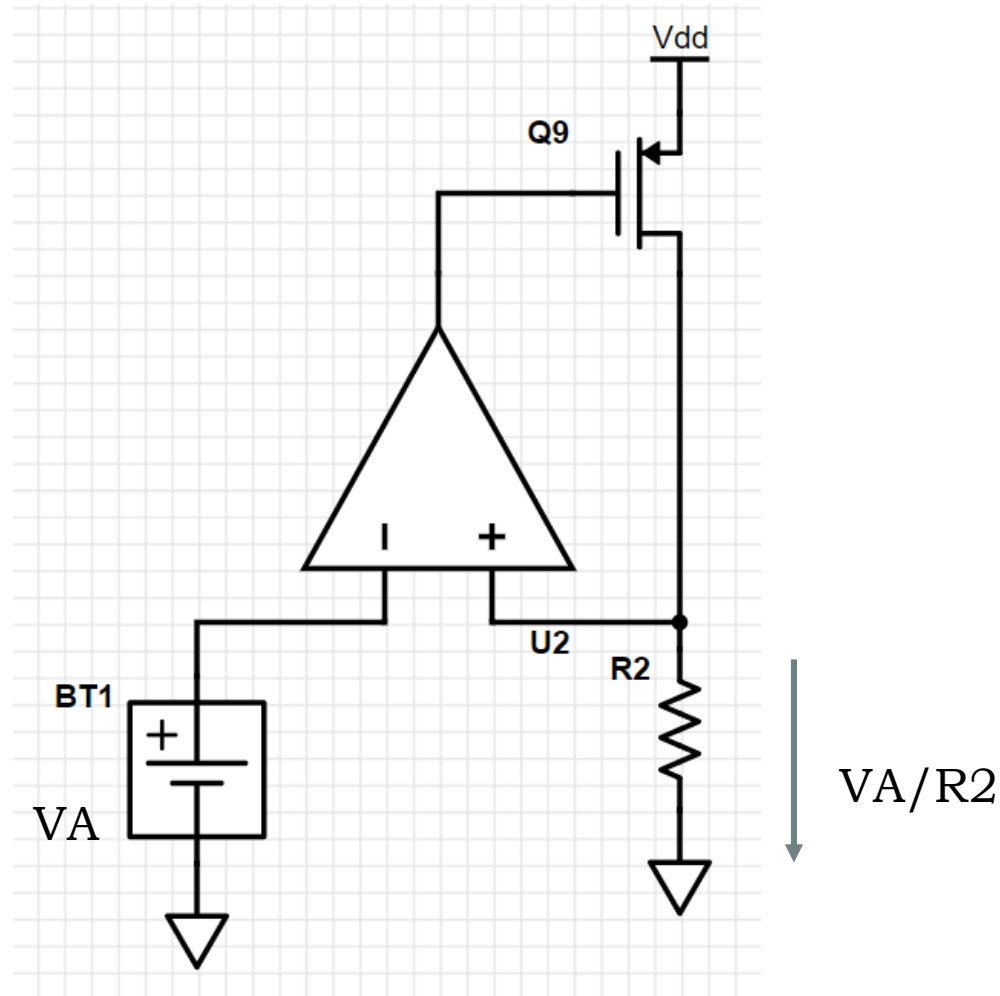
$$MAPV = \frac{\Delta V_{REF}}{V_{REF_{nom}}} (\%)$$

(Smaller the better)

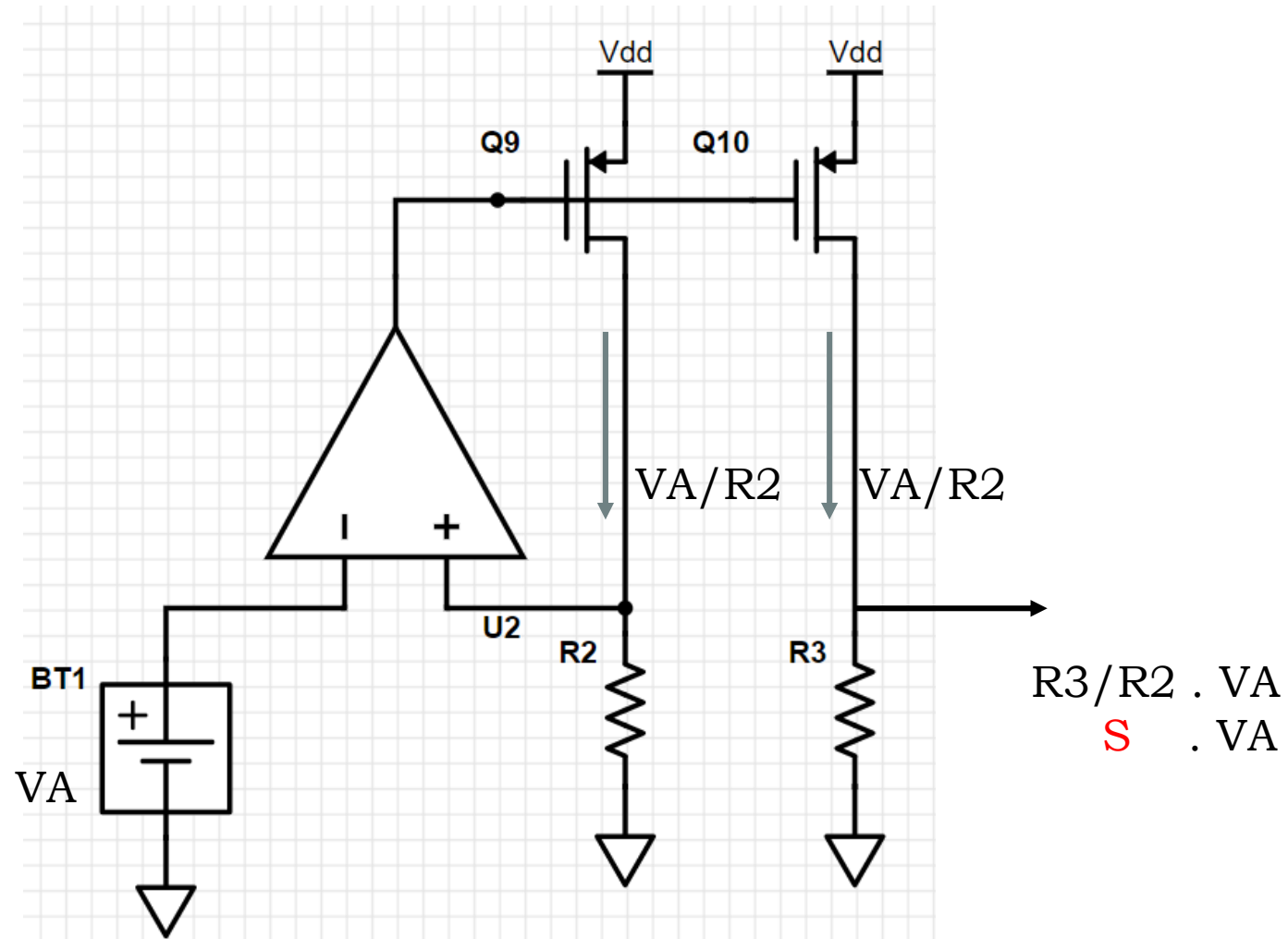
Basic design concept



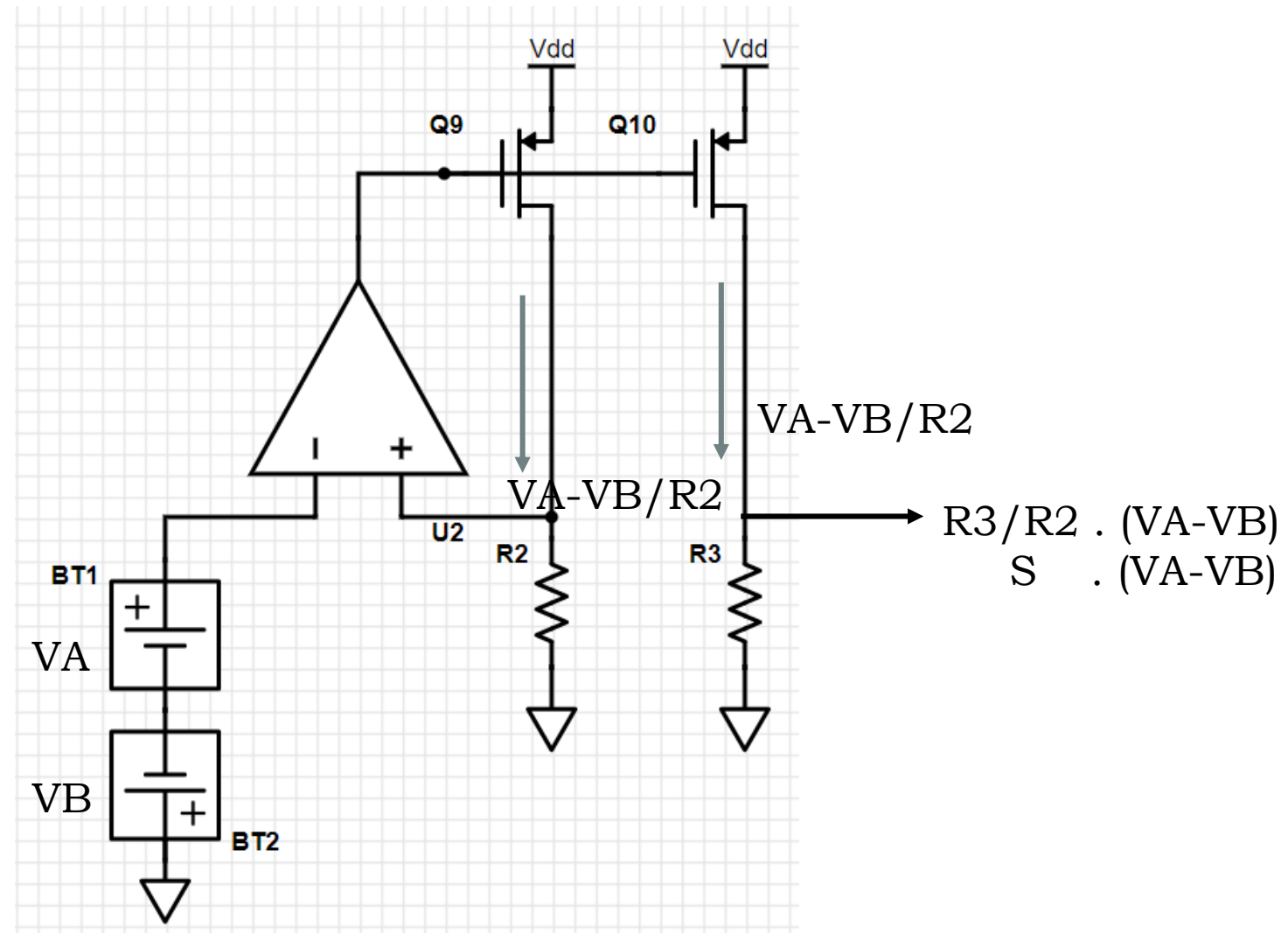
Basic design concept realization



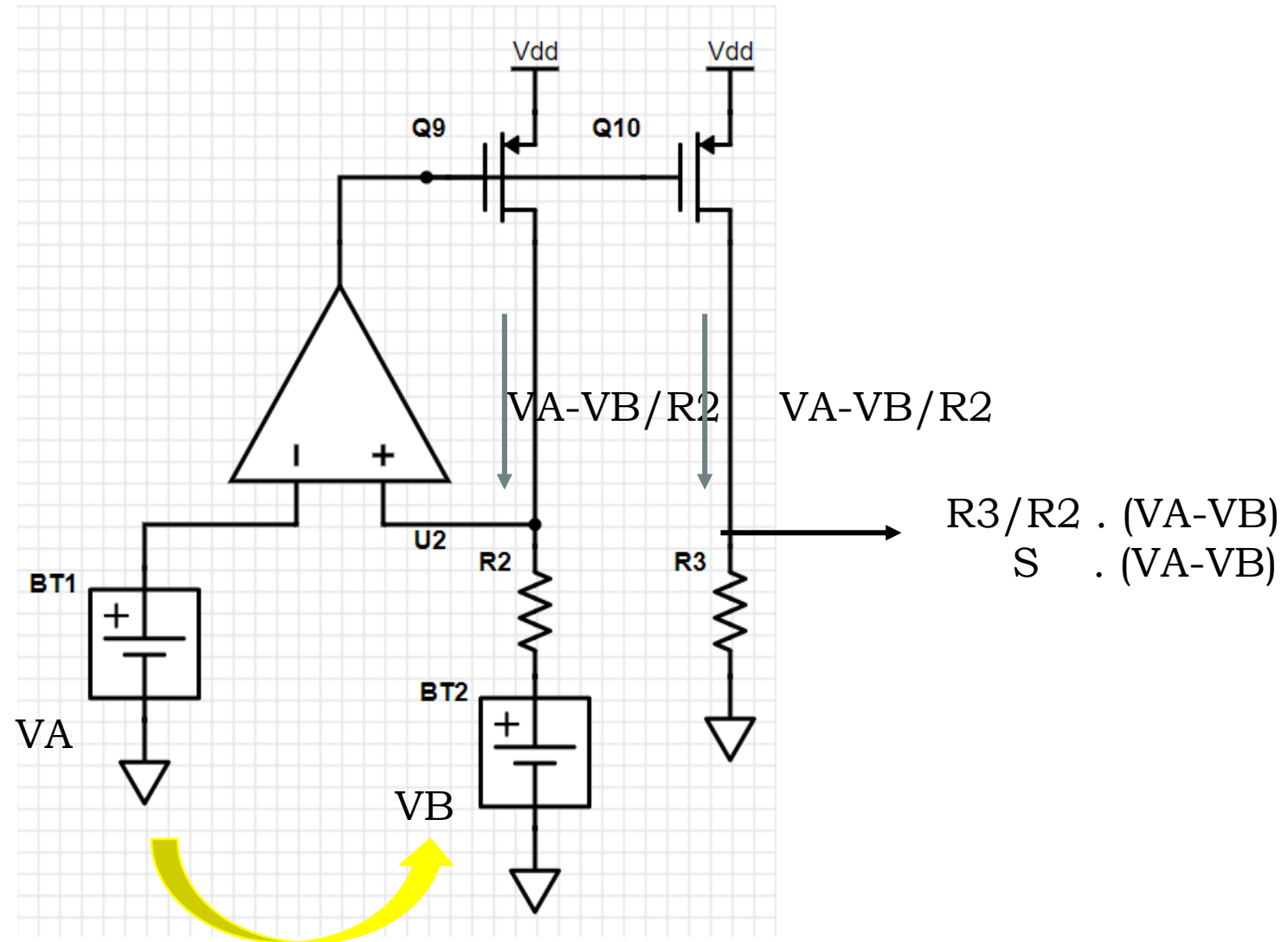
Basic design concept realization



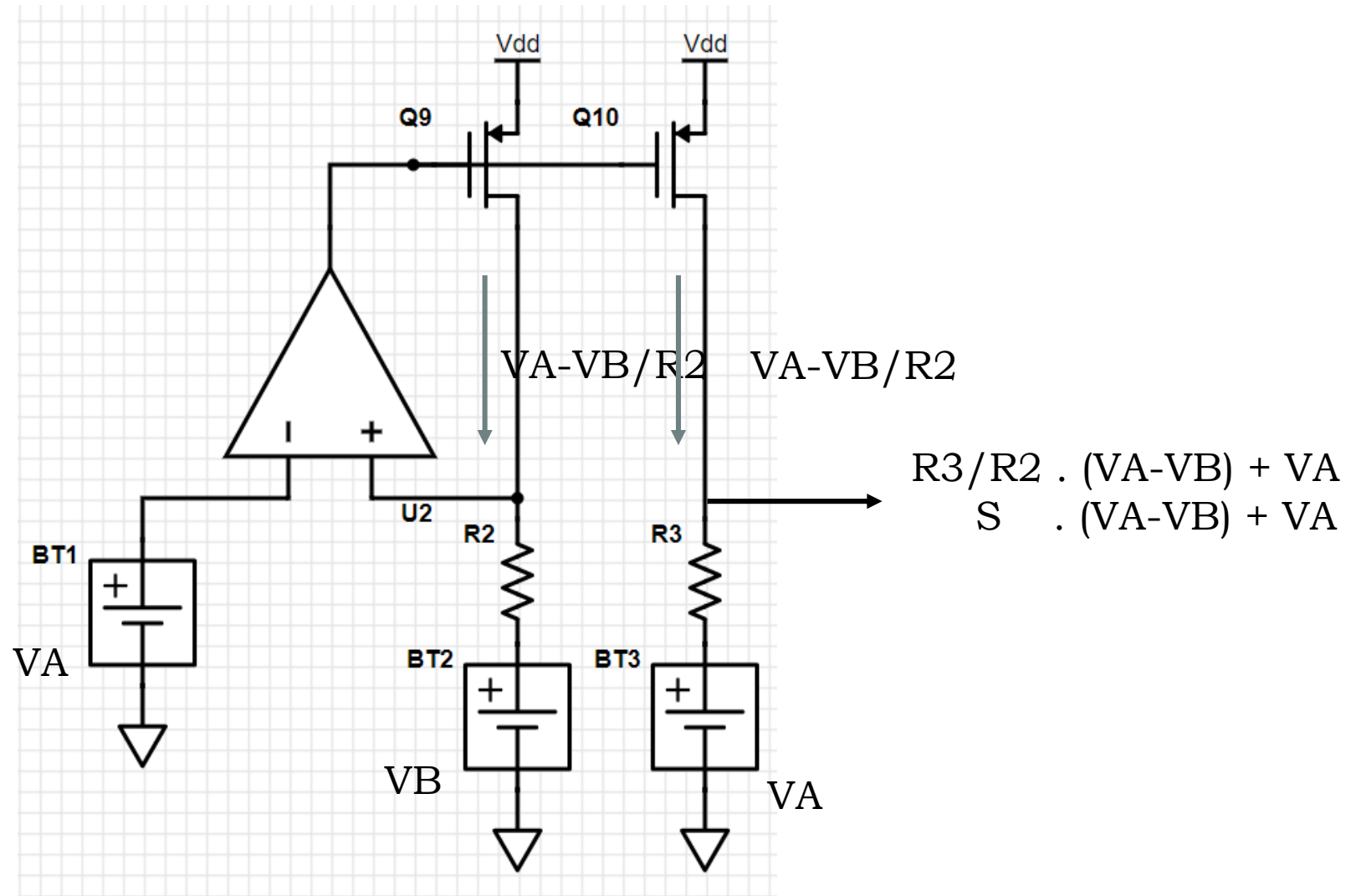
Basic design concept realization



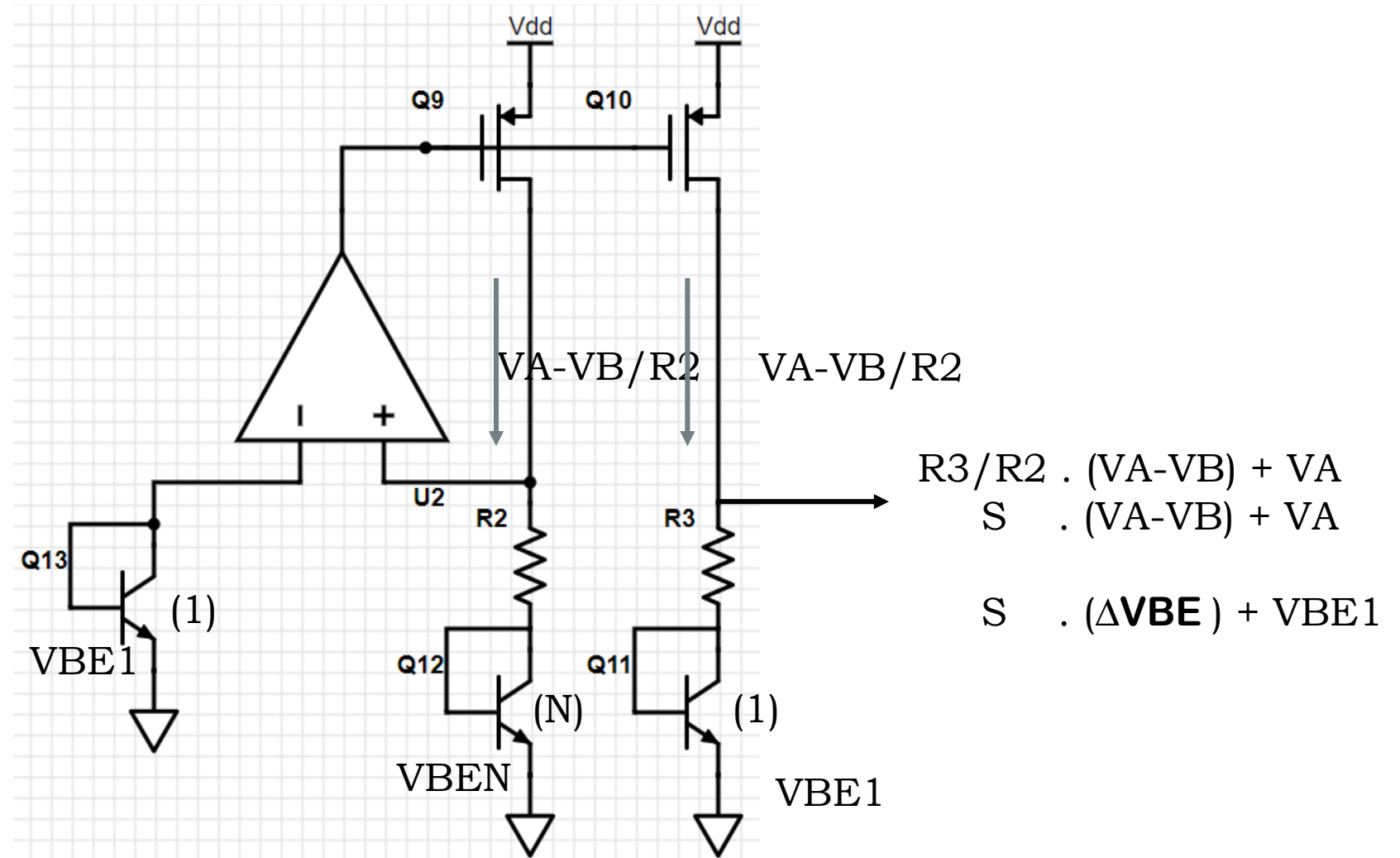
Basic design concept realization



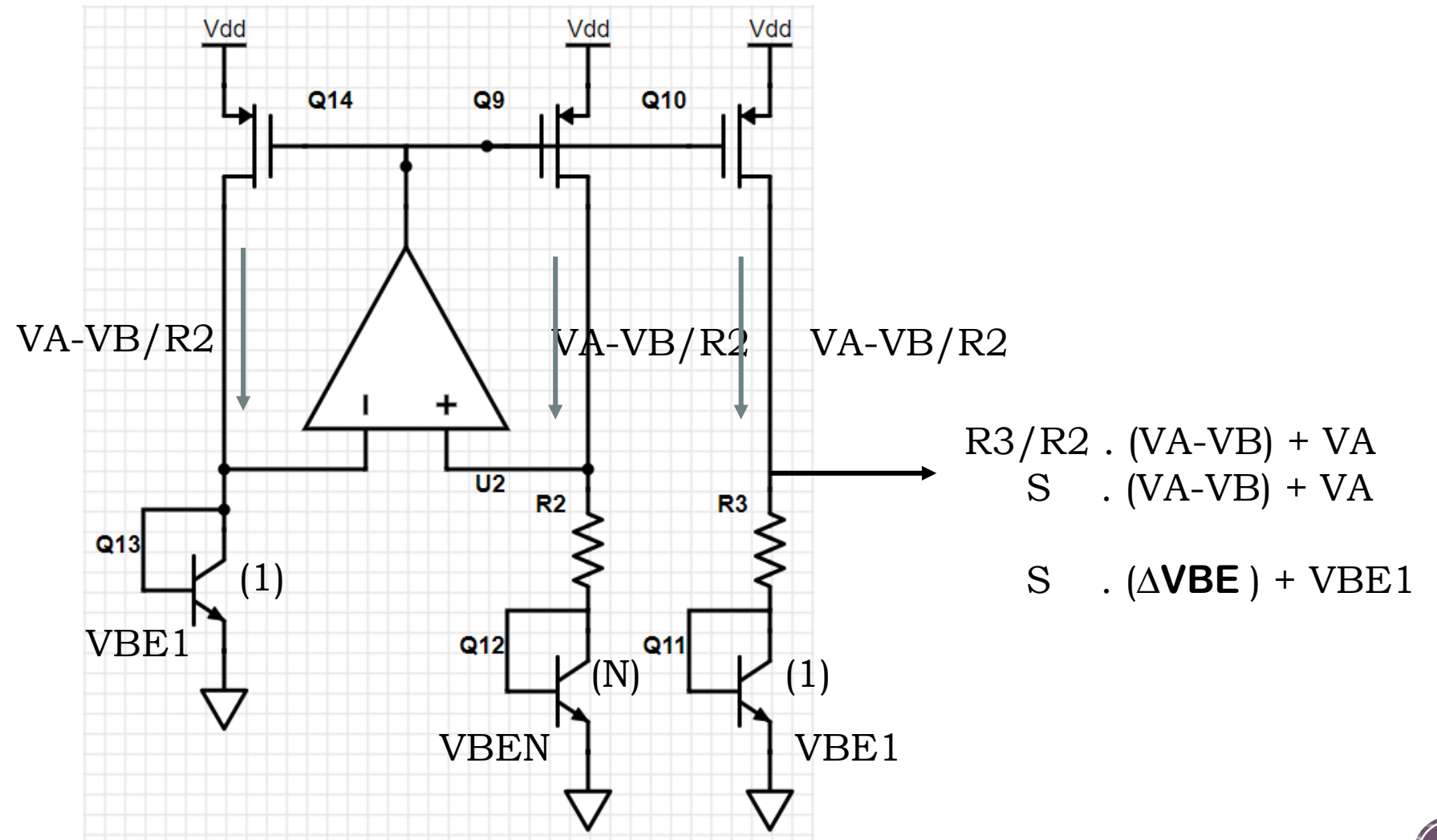
Basic design concept realization



Basic design concept realization



Basic design concept realization



Another Approach !!

Basic design concept realization

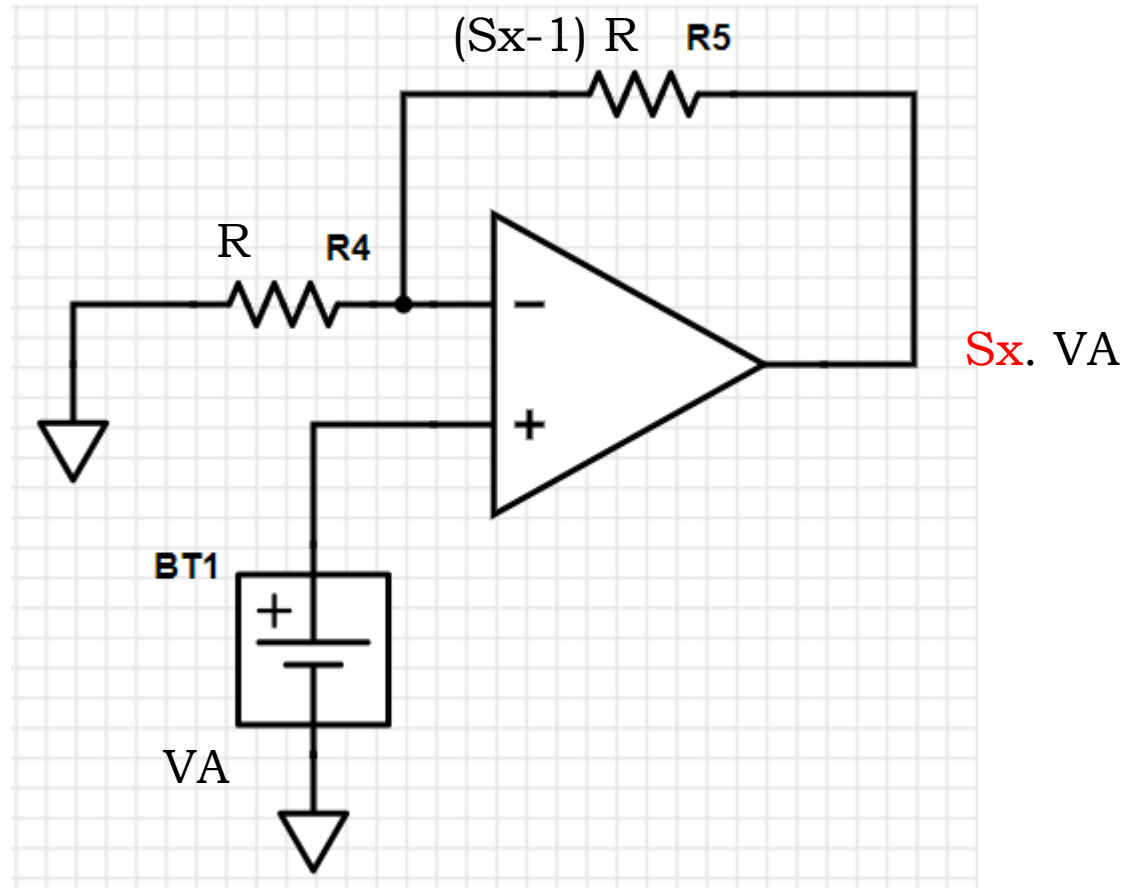
$$V_{BE1} + S \cdot \Delta V_{BE} = V_{REF}$$

$$V_{BE1} = V_A$$

$$V_{BE2} = V_B$$

$$V_A + S \cdot (V_A - V_B)$$

$$V_A + S \cdot V_A - S \cdot V_B$$



Basic design concept realization

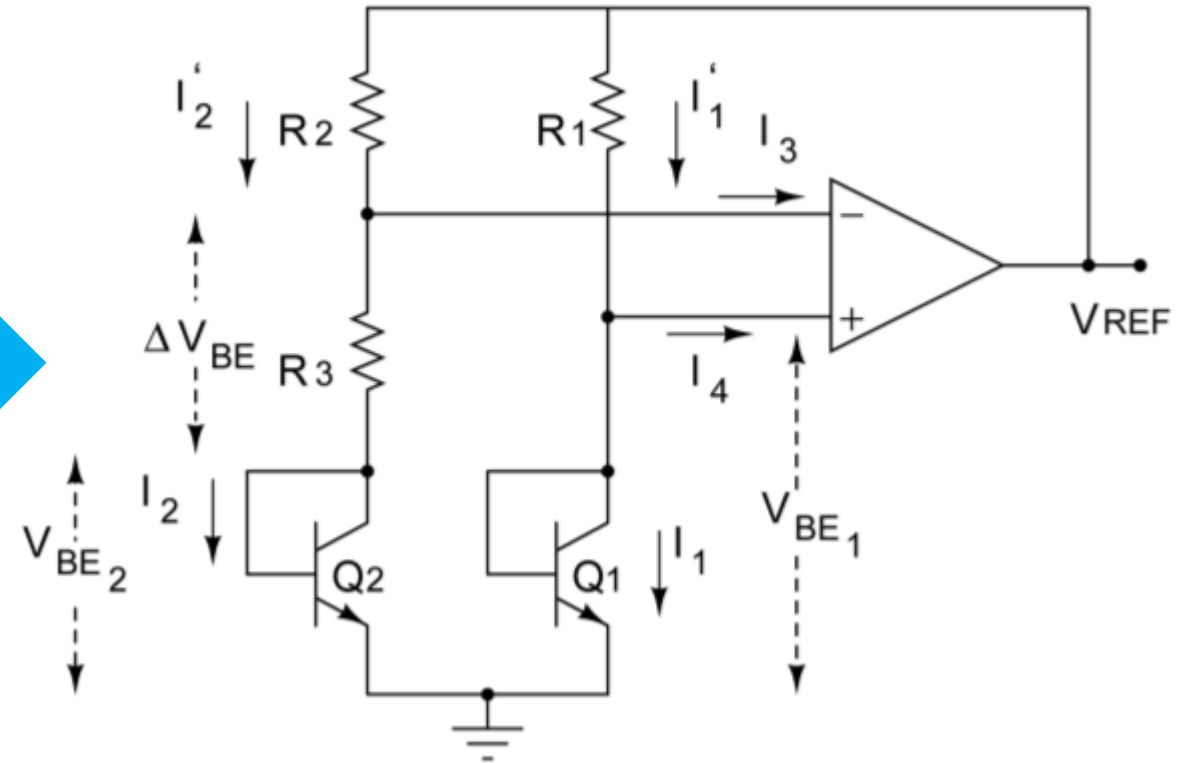
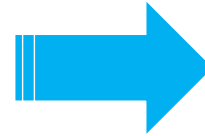
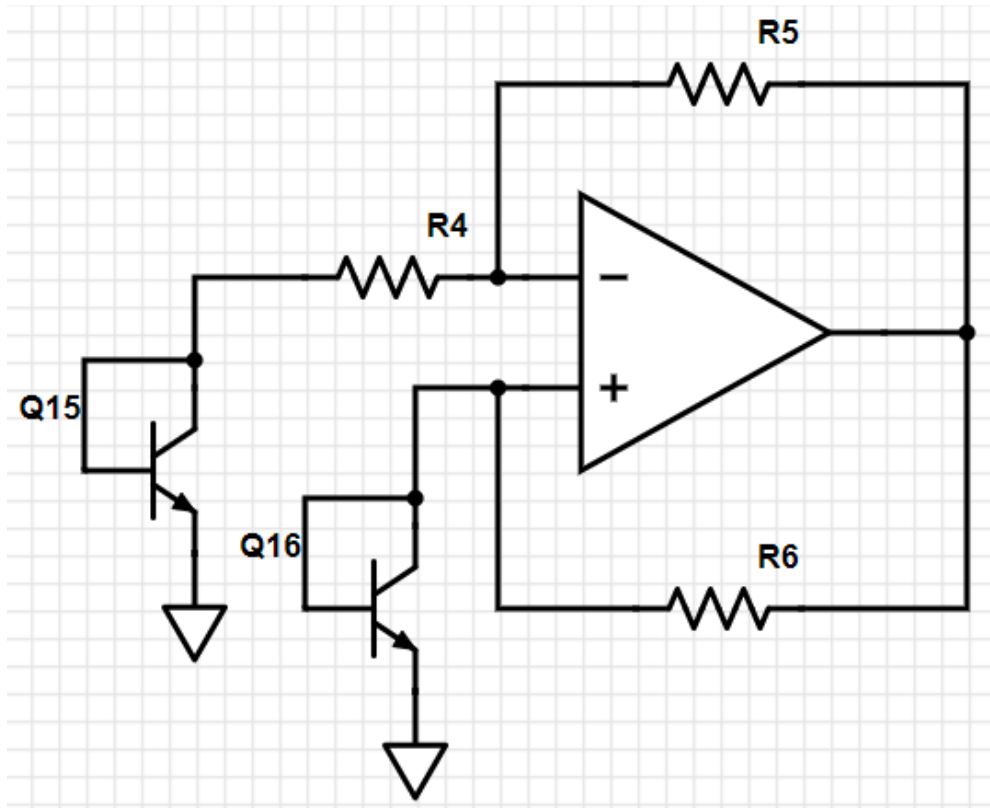


Figure . Kuijk voltage reference: 1973.

Requirement !!

Requirement

$$V_{BE1} \approx V_T \ln (I/I_s)$$

$$V_{BE1} = V_T \ln [(I/I_s) + 1]$$

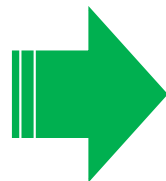
$$V_{BE2} = V_T \ln [(I/N.I_s) + 1]$$

$$\Delta V_{BE} = V_T \ln [N.(I+I_s)/(I+N.I_s)]$$

$$I = \Delta V_{BE}/R$$

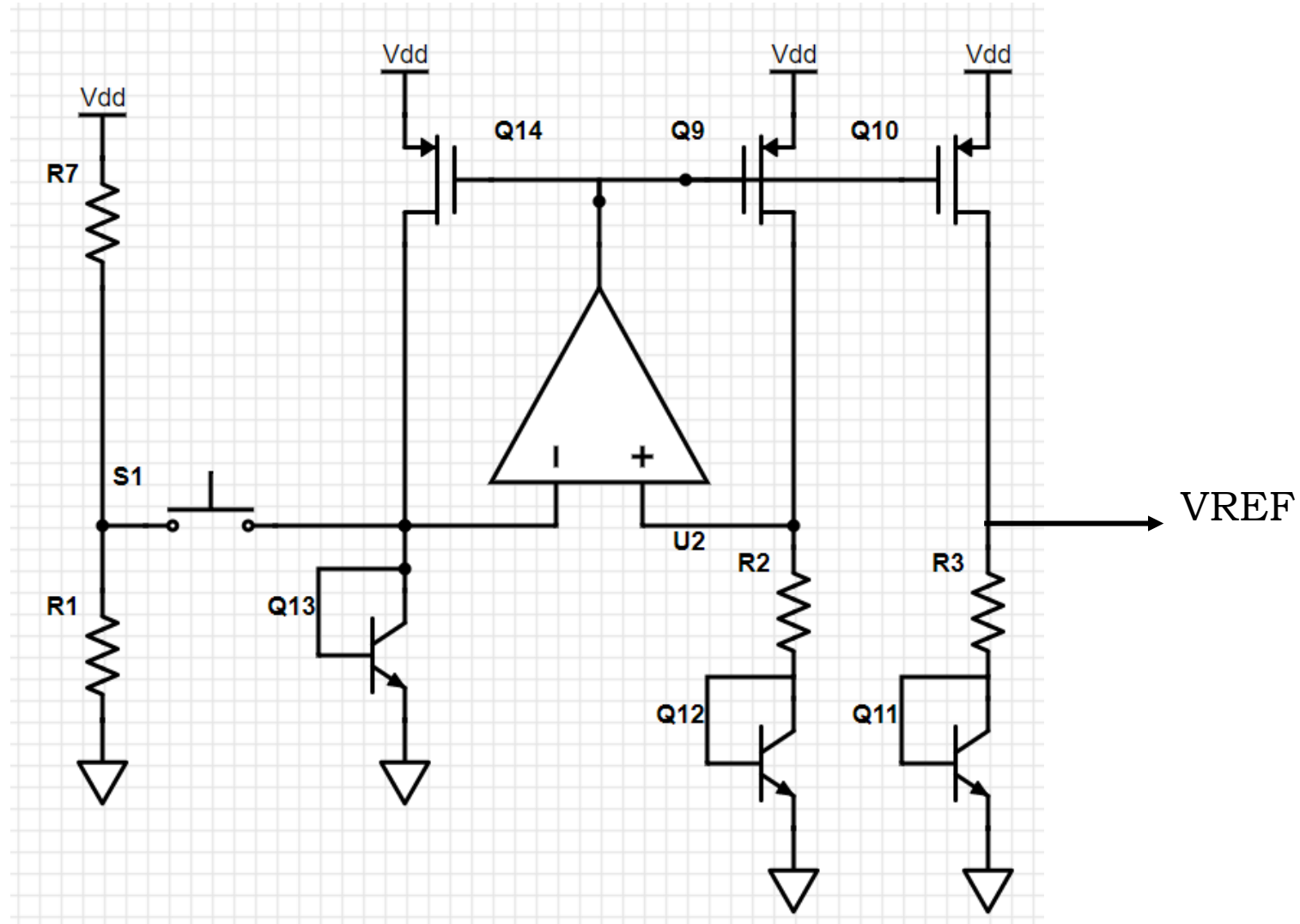
$$= V_T/R \ln [N.(I+I_s)/(I+N.I_s)]$$

2 Solutions !!!!



$I = 0$ and $I = \Delta V_{BE}/R$

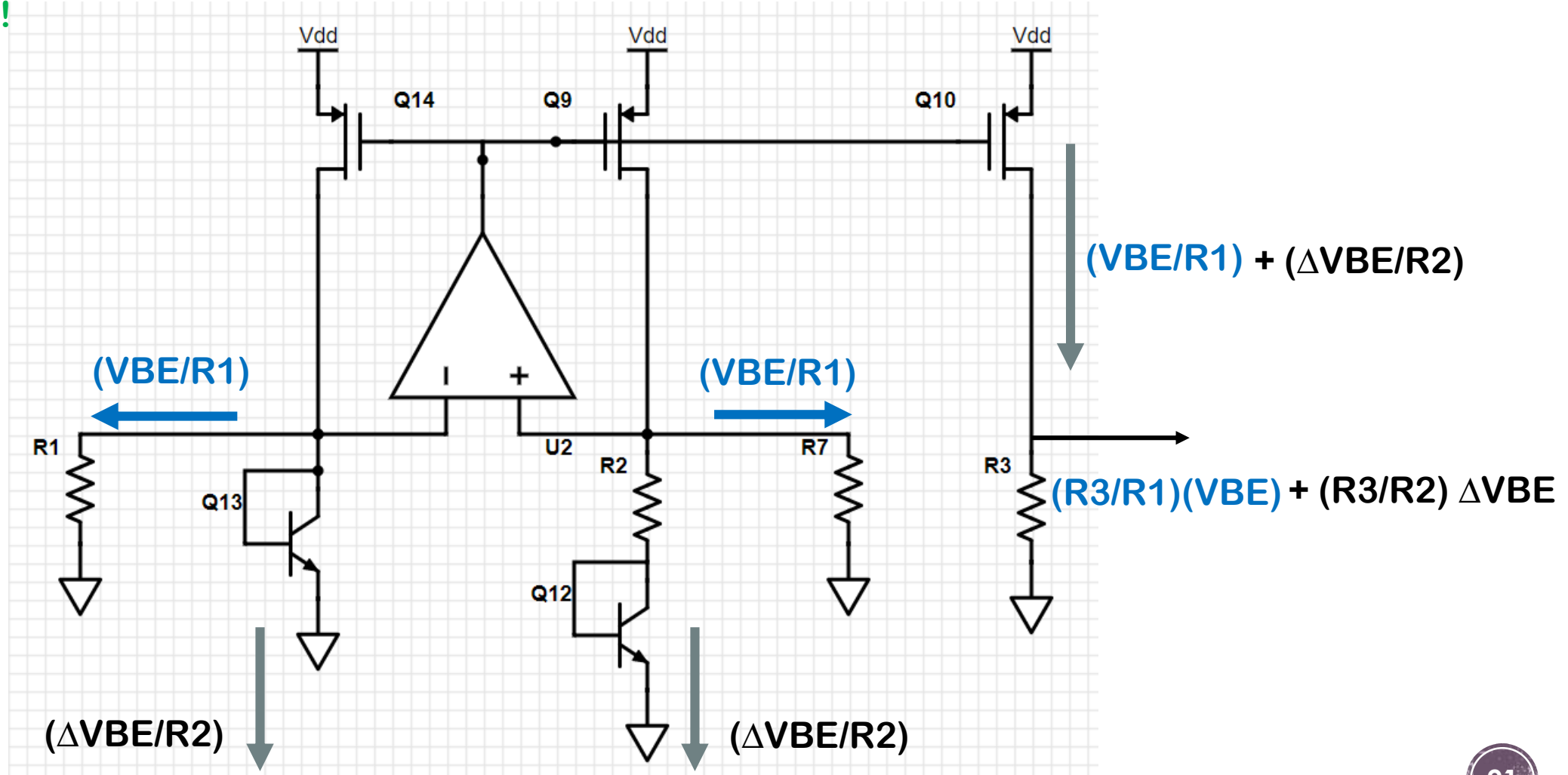
Start-up requirement



Popular Sub-1V BGR !!!

Popular Sub-1V BGR : Banba et al. JSSC'99

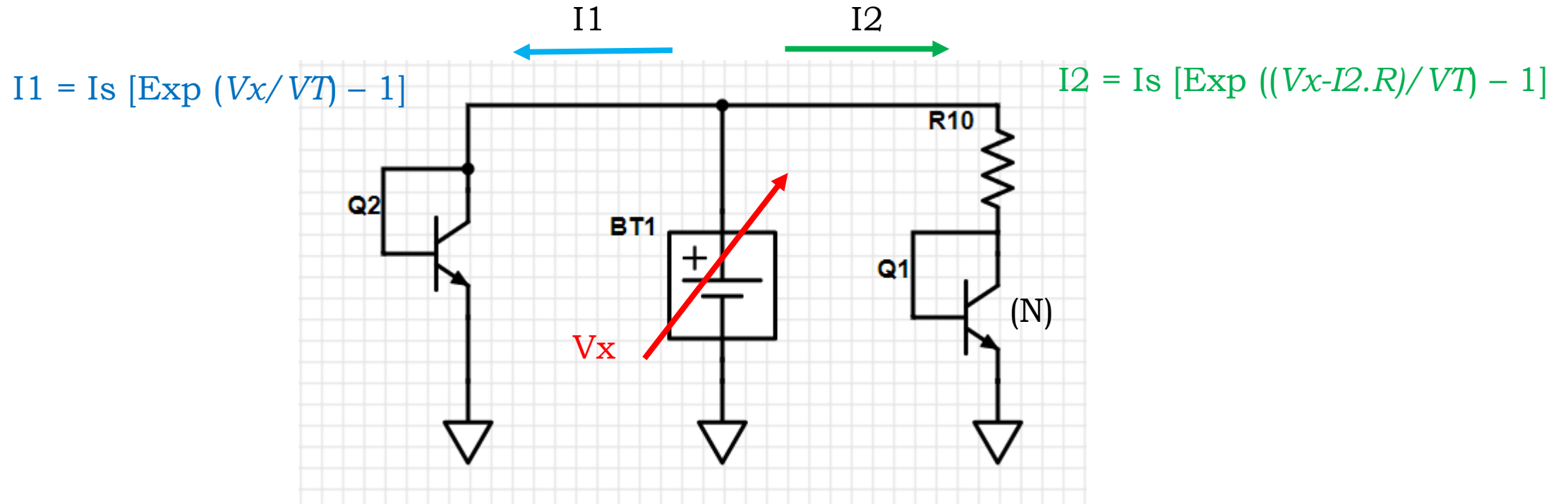
Current Mode !!!



One MORE Approach for BGR !!

Basic design concept realization

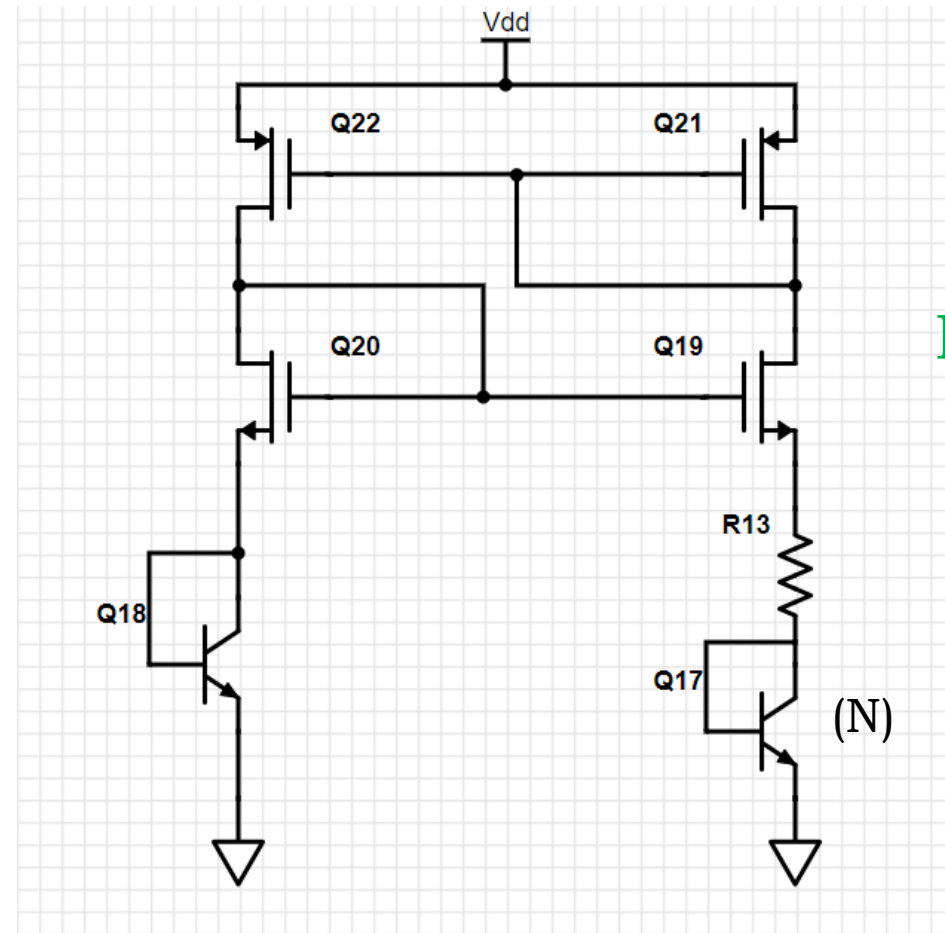
We have to make $I_1 = I_2 = I = \Delta V_{BE}/R$



$I_1 < I_2$ ----- V_x is small ----- V_x needs to be increased !!!

$I_2 < I_1$ ----- V_x is large ----- V_x needs to be decreased !!!

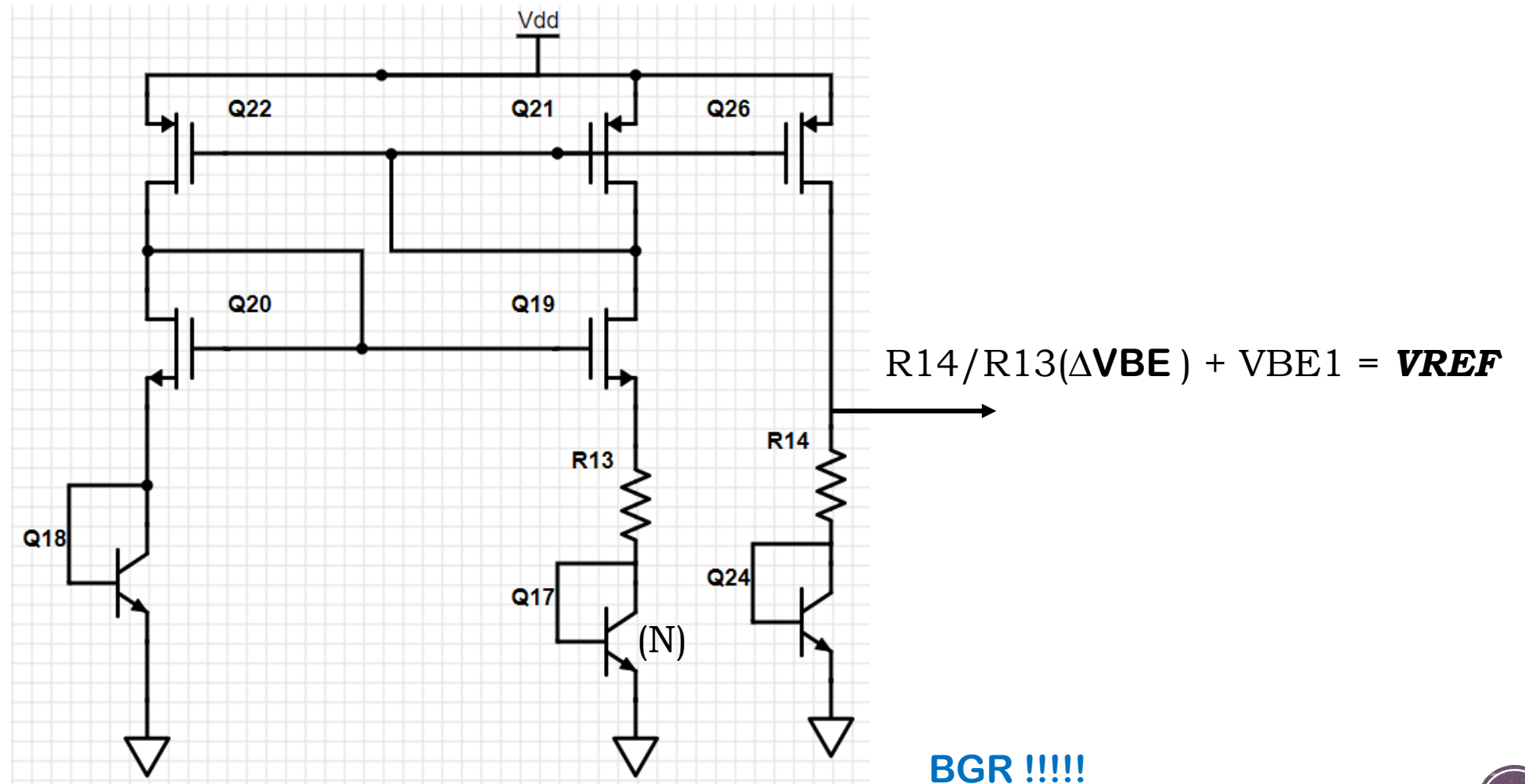
Basic design concept realization



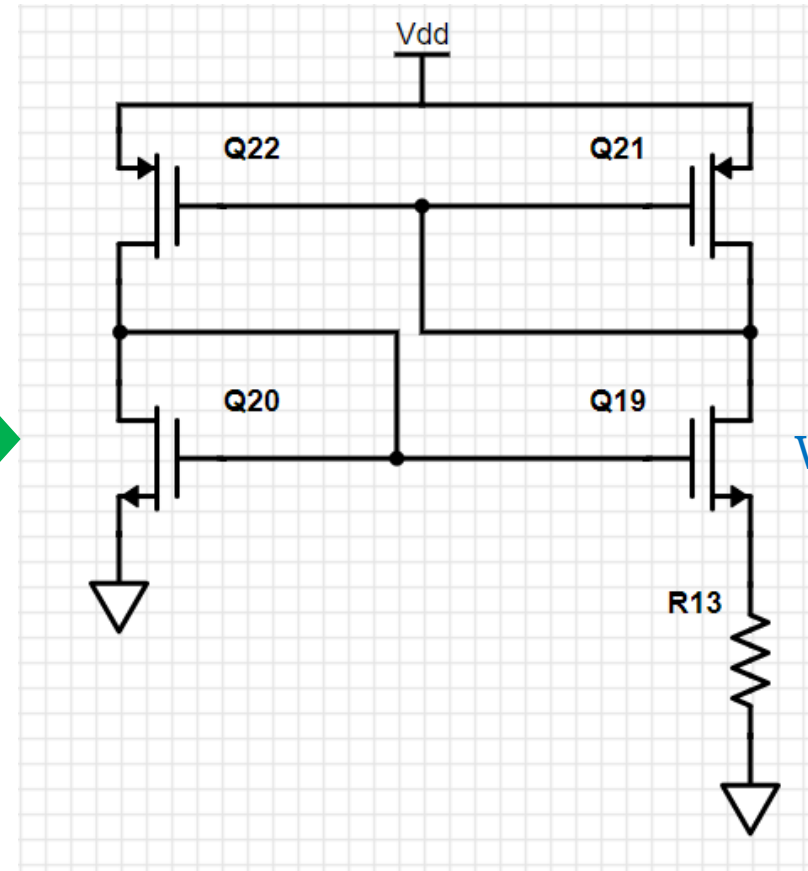
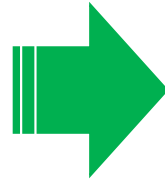
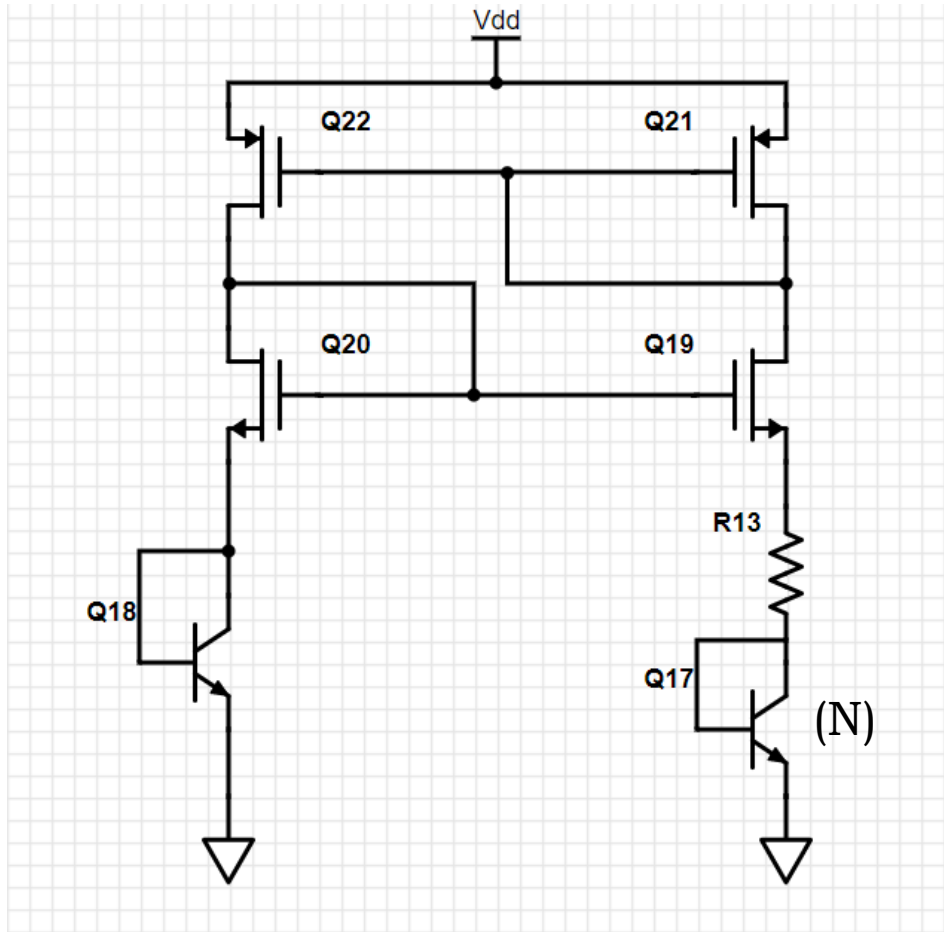
$$I_1 = I_2 = I$$

We have to make $I_1 = I_2 = I = \Delta V_{BE}/R$ ---- PTAT Current !!!!!

Basic design concept realization

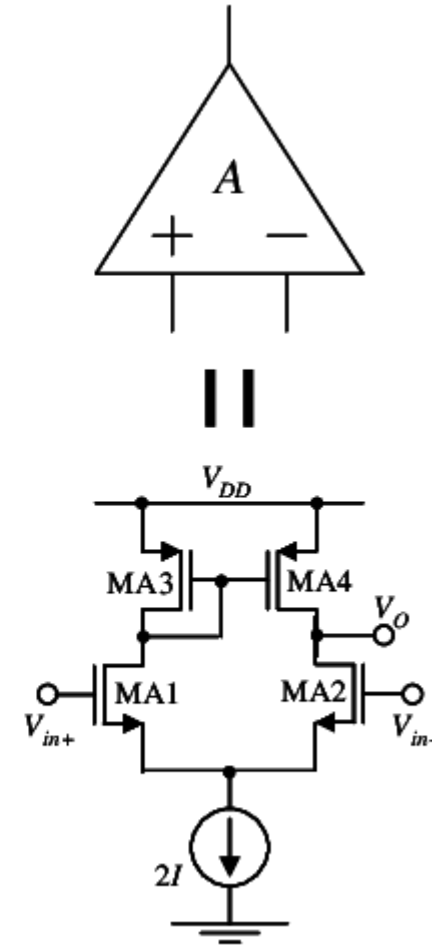


Derived Constant-gm Bias circuit

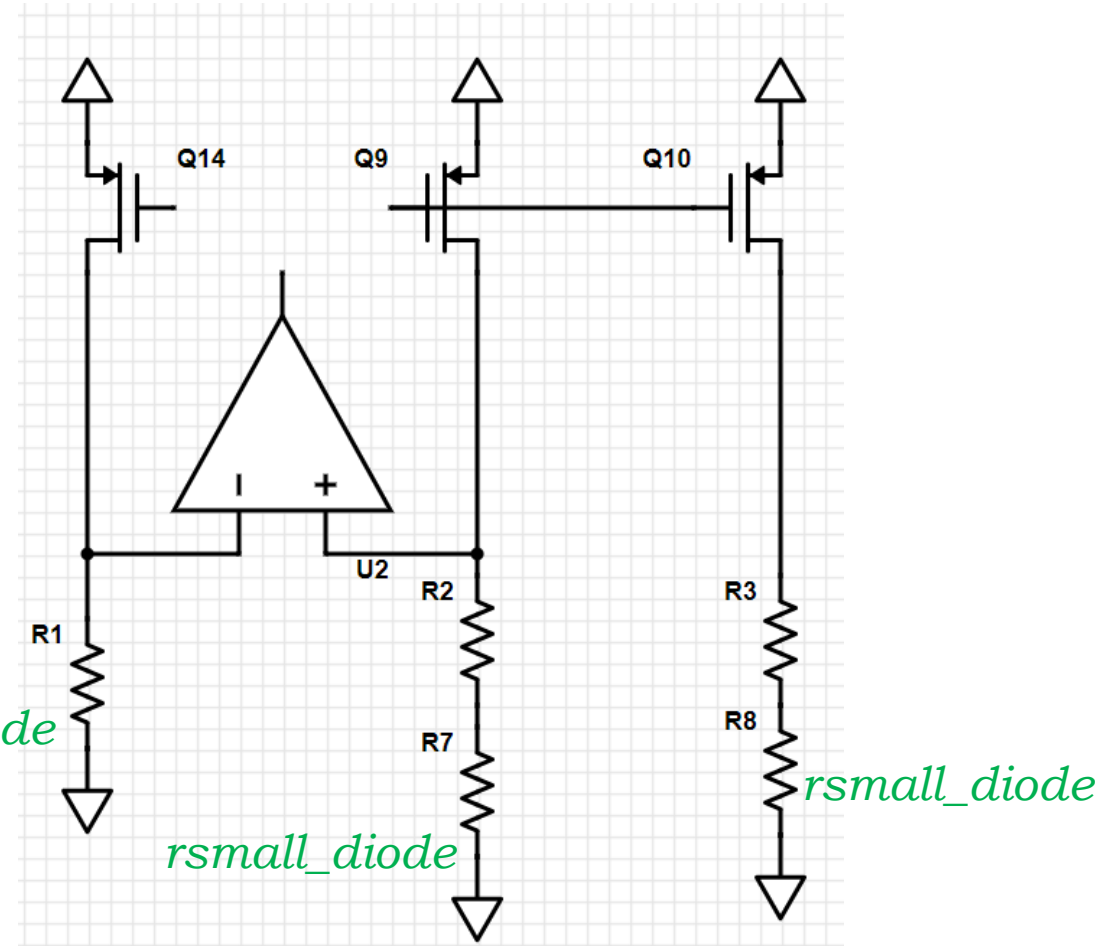
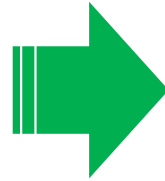
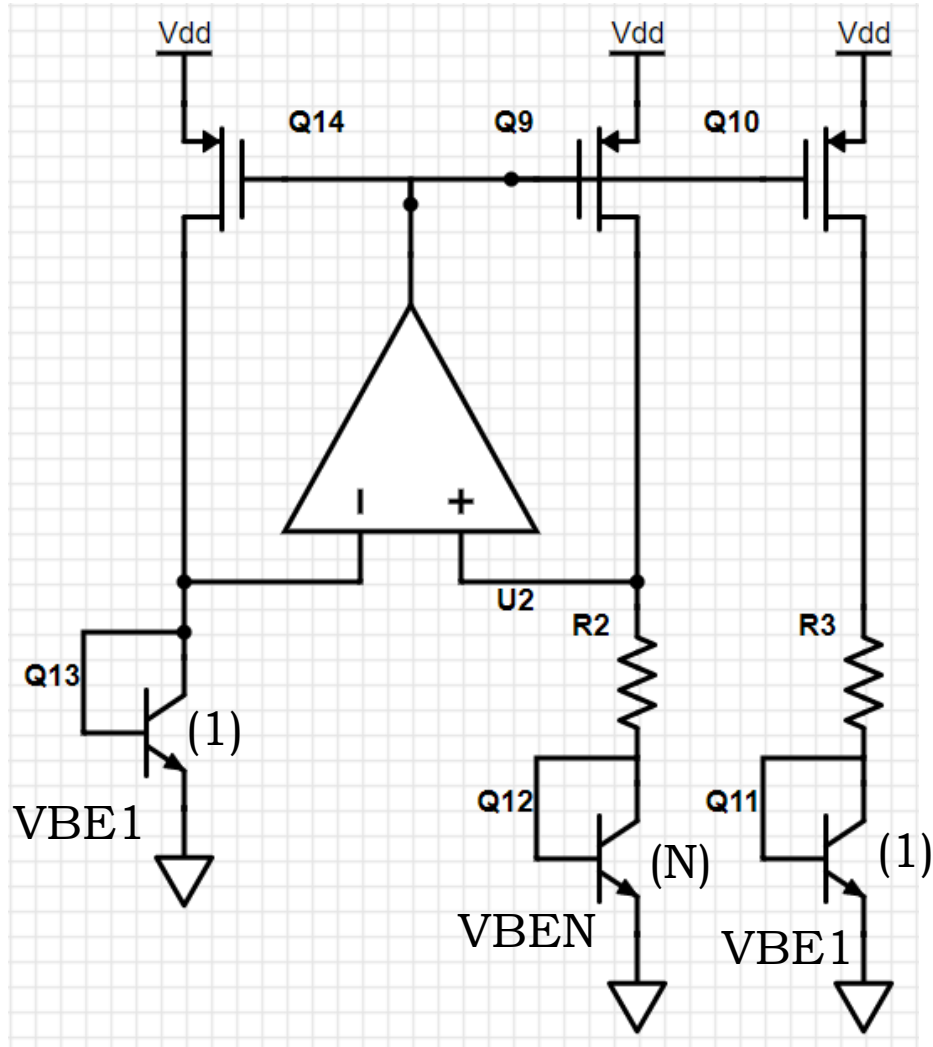


Width: K times more

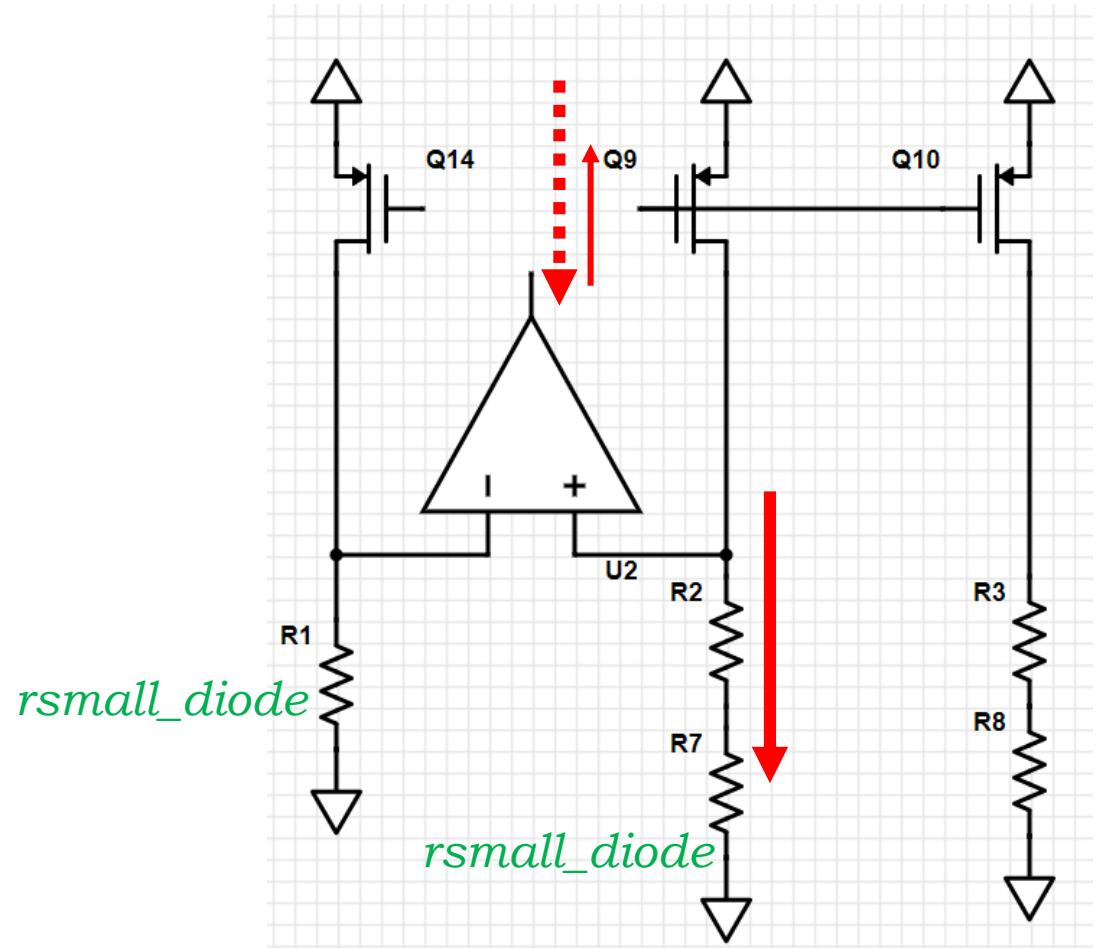
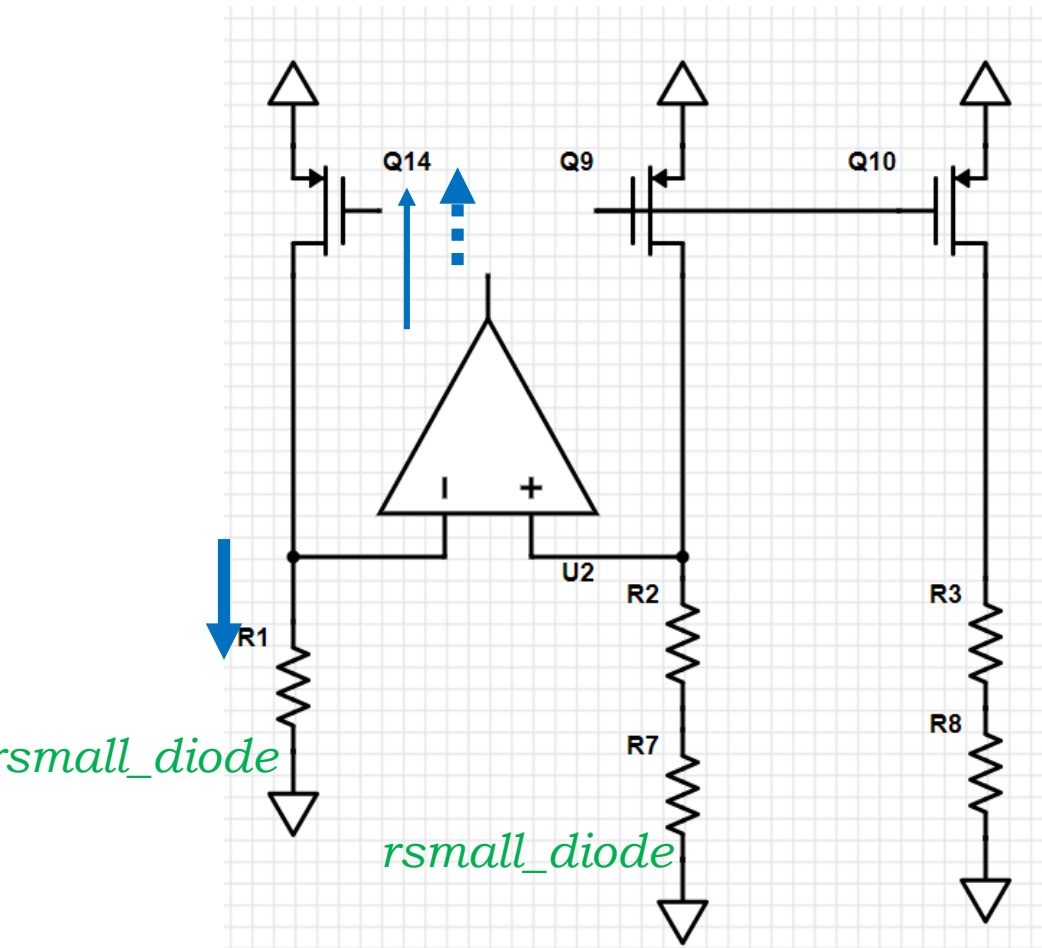
Things to remember!!



Small Signal Analysis



Small Signal Analysis



Positive Feedback **WEAKER than** Negative Feedback

Optimisation !!!

Optimisation

$$V_{REF} = S \cdot (\Delta V_{BE}) + V_{BE1}$$

$$V_{REF} = R2/R1 \cdot [V_T \ln(N)] + V_{BE1}$$

$$dV_{REF}/dT = R2/R1 \cdot [k/q \ln(N)] - 2m$$

AND/OR

$$dV_{REF}/dT = R2/R1 \cdot [k/q \ln(I2/I1 \cdot N)] - 2m$$

Optimisation

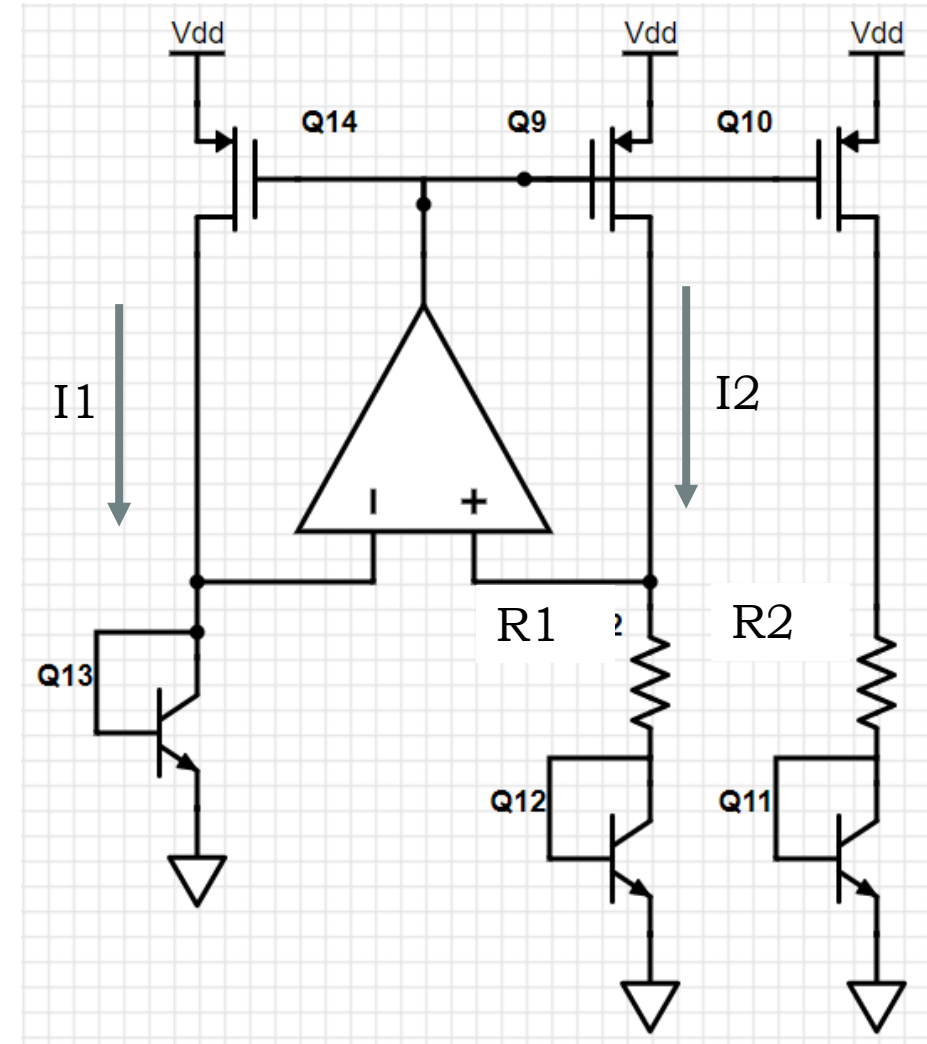
$$V_{REF} = S \cdot (\Delta V_{BE}) + V_{BE1}$$

$$V_{REF} = R2/R1 \cdot [V_T \ln(N)] + V_{BE1}$$

$$dV_{REF}/dT = R2/R1 \cdot [k/q \ln(N)] - 2m$$

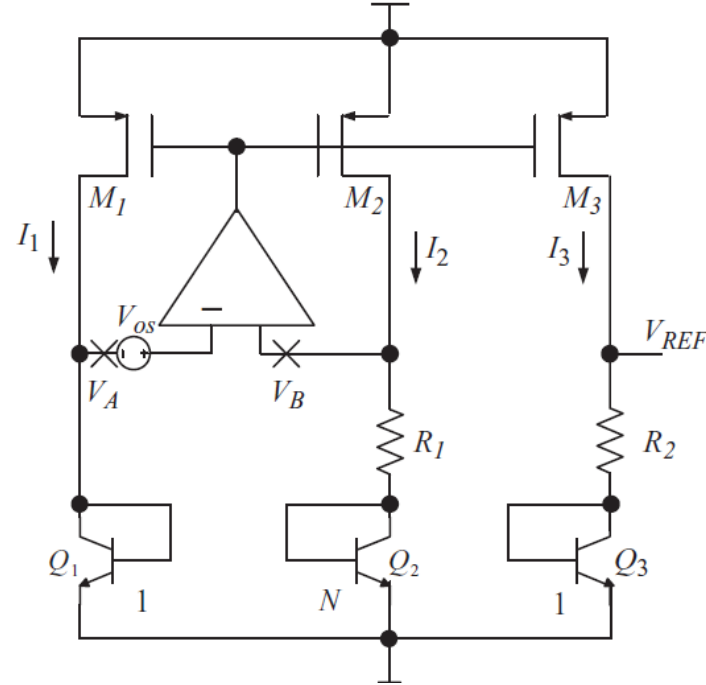
AND/OR

$$dV_{REF}/dT = R2/R1 \cdot [k/q \ln(I2/I1 \cdot N)] - 2m$$



Error sources in voltage references !!!

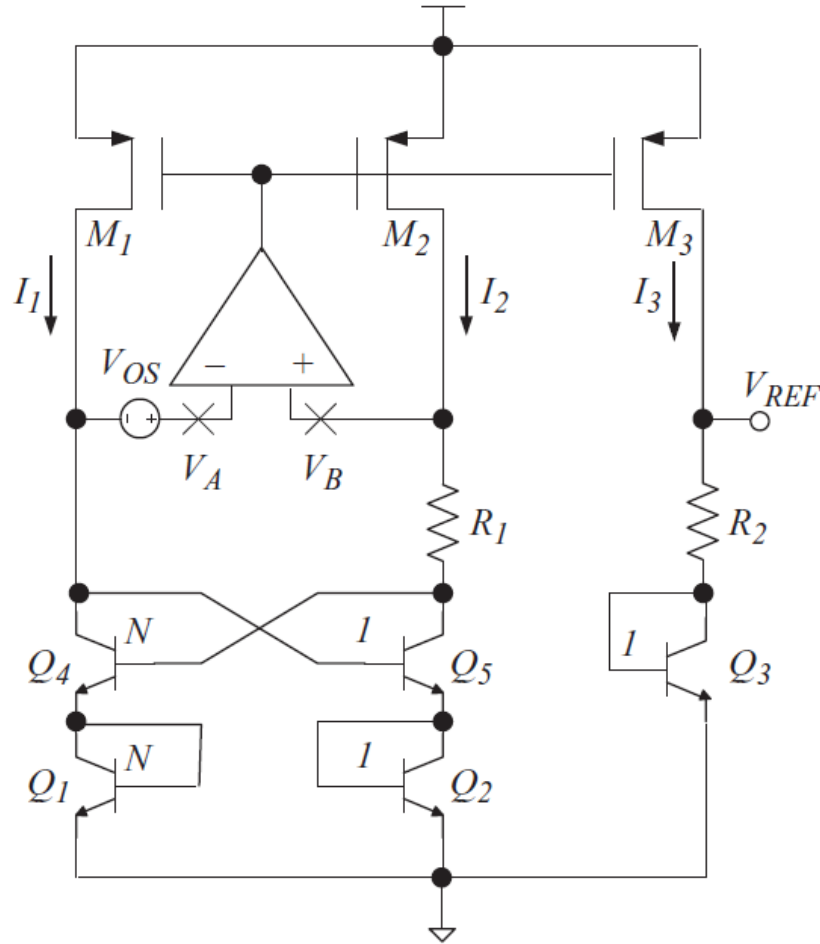
Error source : OFFSET



$$V_{REF} = \frac{R_2}{R_1} V_T \ln N + V_{BE_3} + \frac{R_2}{R_1} V_{OS}$$

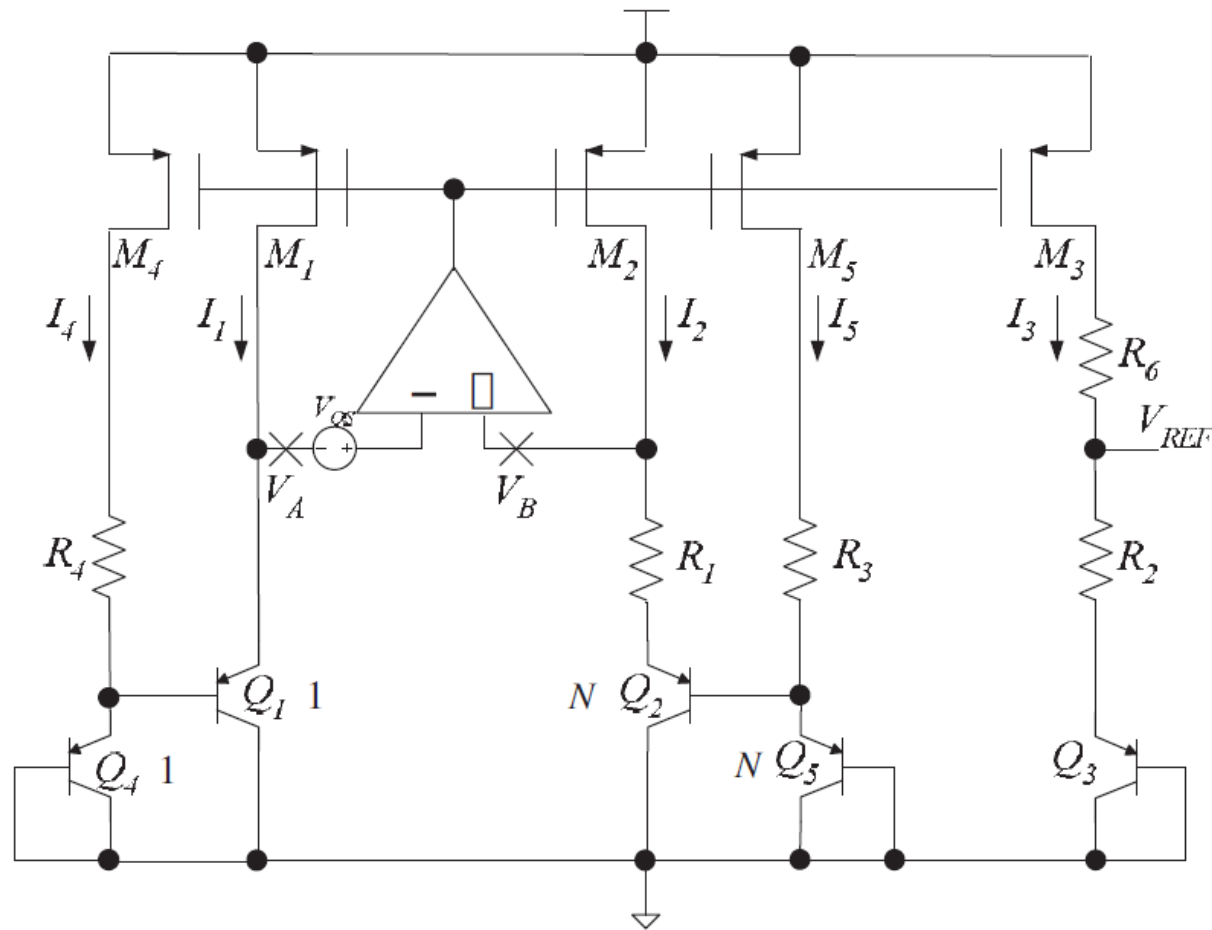
$$= V_{REF-CONV} + \Delta V_{REF,OS},$$

Error source : OFFSET solution



$$\begin{aligned}
 V_{REF} &= I_3 R_2 + V_{BE_3} \\
 &= \frac{R_2}{R_1} \ln(N^2) V_T + V_{BE_3} + \frac{R_2}{R_1} V_{OS} \\
 &= V_{REF-CONV} + \Delta V_{REF,OS},
 \end{aligned}$$

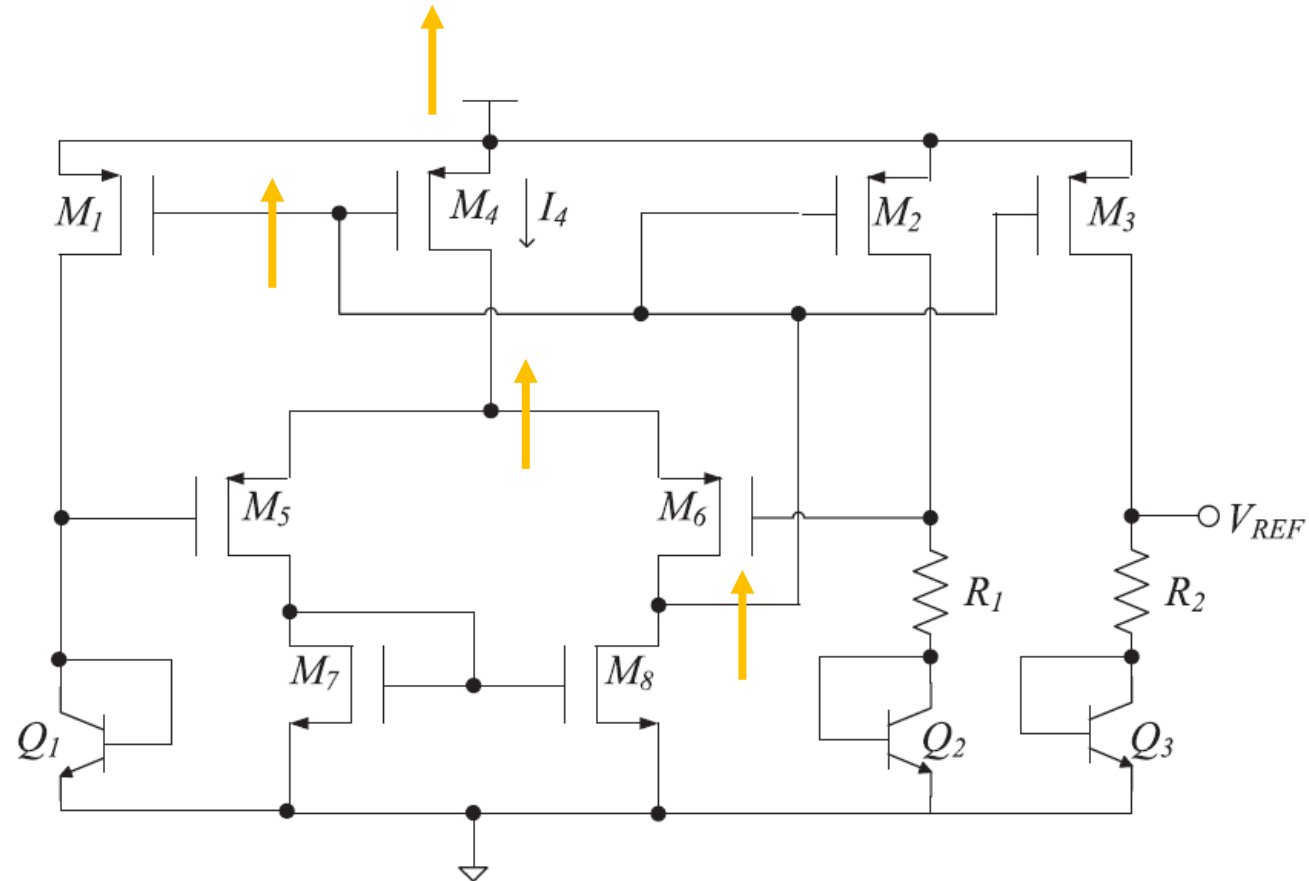
Error source : OFFSET solution



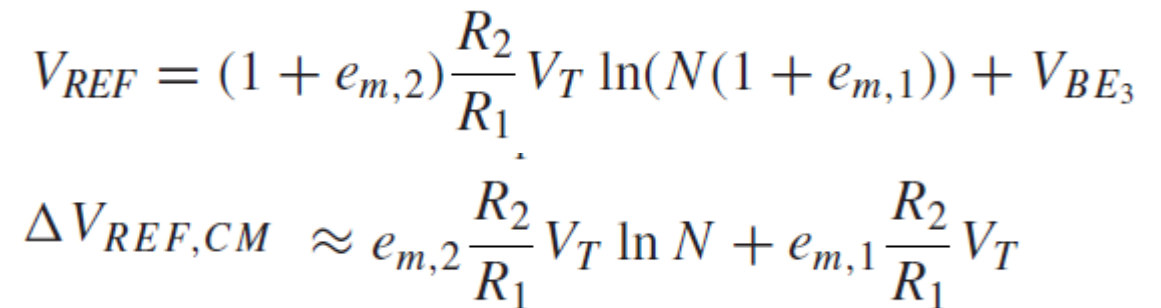
$$\begin{aligned}
 V_{REF} &= V_{EB3} + I_3 R_2 \\
 &= V_{EB3} + \frac{I_2}{2} R_2 \\
 &= V_{EB3} + \frac{1}{2} \frac{R_2}{R_1} (2V_T \ln(N) + V_{OS}) \\
 &= V_{EB3} + \frac{R_2}{R_1} V_T \ln(N) + \frac{1}{2} \frac{R_2}{R_1} V_{OS} \\
 &= V_{REF-CONV} + \Delta V_{REF,OS},
 \end{aligned}$$

Error source : PSRR solution

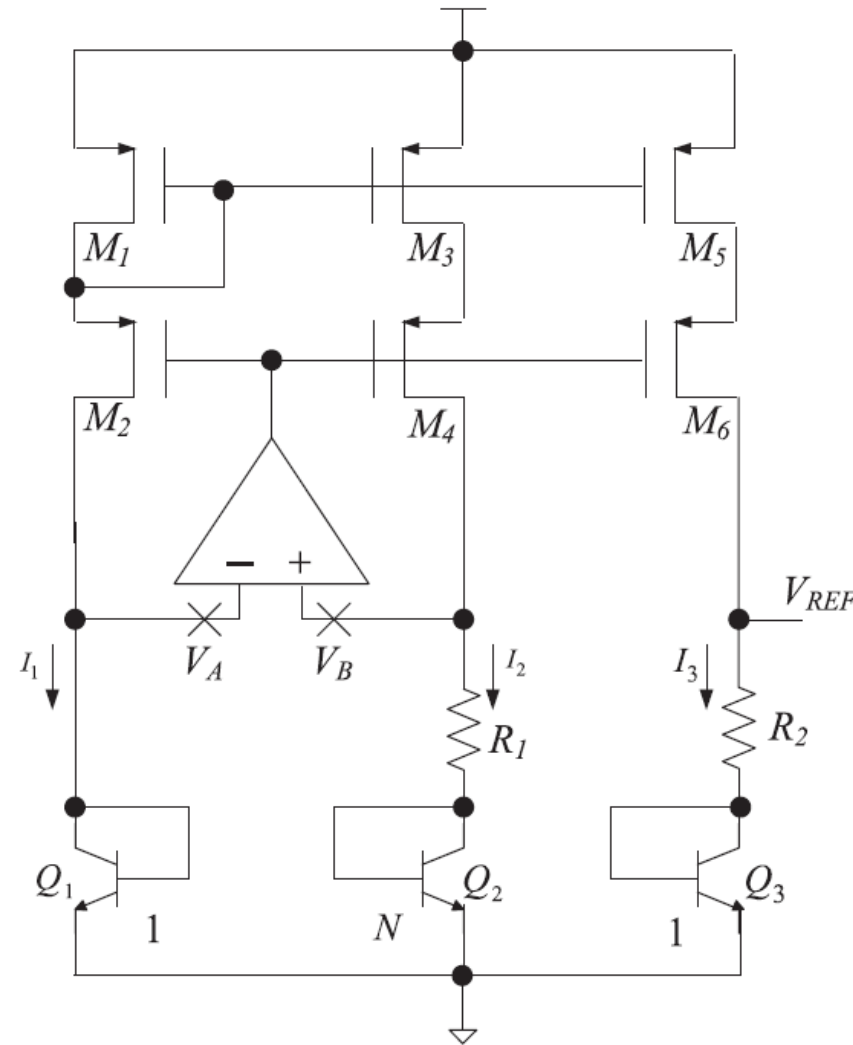
Tracking



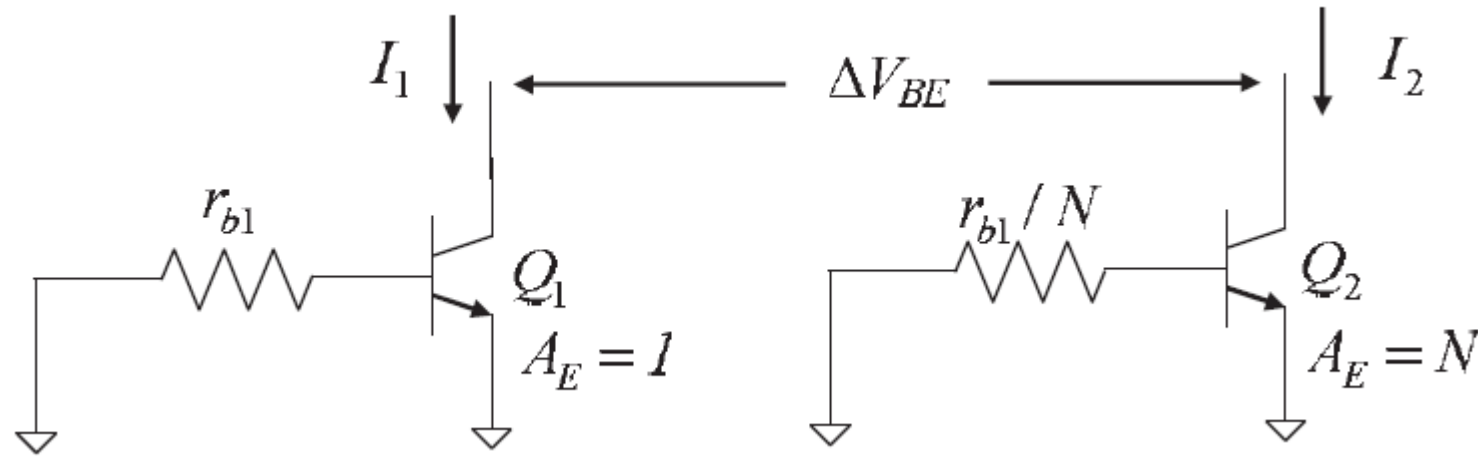
Systematic Mismatch



Error sources : Current Mismatch , Solution



Error sources : Base resistance



$$V_{REF} = \frac{R_2}{R_1} V_T \ln N + V_{BE_{30}} + \Delta V_{REF, r_b}$$

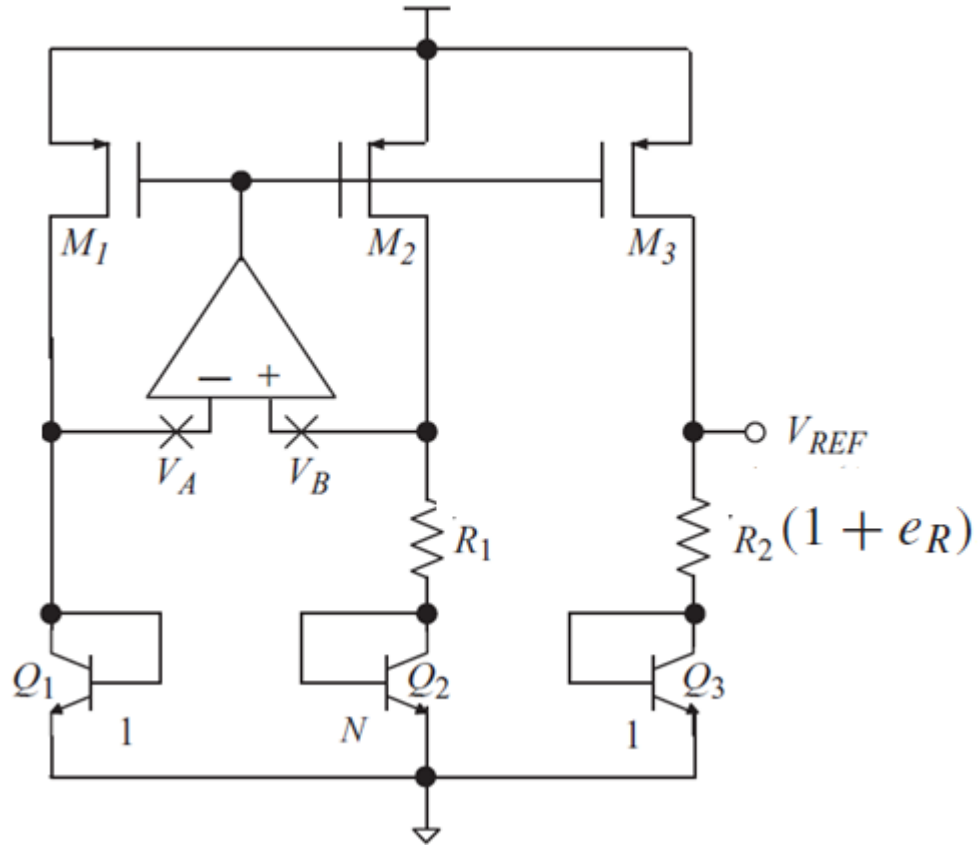
$$\Delta V_{REF, r_b} \approx \frac{R_2}{R_1} V_T \ln \left(N \frac{\Delta r_{b1,2}}{\beta R_1} \right) + \frac{R_2}{R_1} V_T \ln \left(N \frac{r_{b3}}{\beta R_2} \right)$$

Error sources : Base resistance, solutions

$$V_{REF} = \frac{R_2}{R_1} V_T \ln N + V_{BE_{30}} + \Delta V_{REF, r_b}$$

$$\Delta V_{REF, r_b} \approx \frac{R_2}{R_1} V_T \ln \left(N \frac{\Delta r_{b1,2}}{\beta R_1} \right) + \frac{R_2}{R_1} V_T \ln \left(N \frac{r_{b3}}{\beta R_2} \right)$$

Error sources : Resistor mismatch

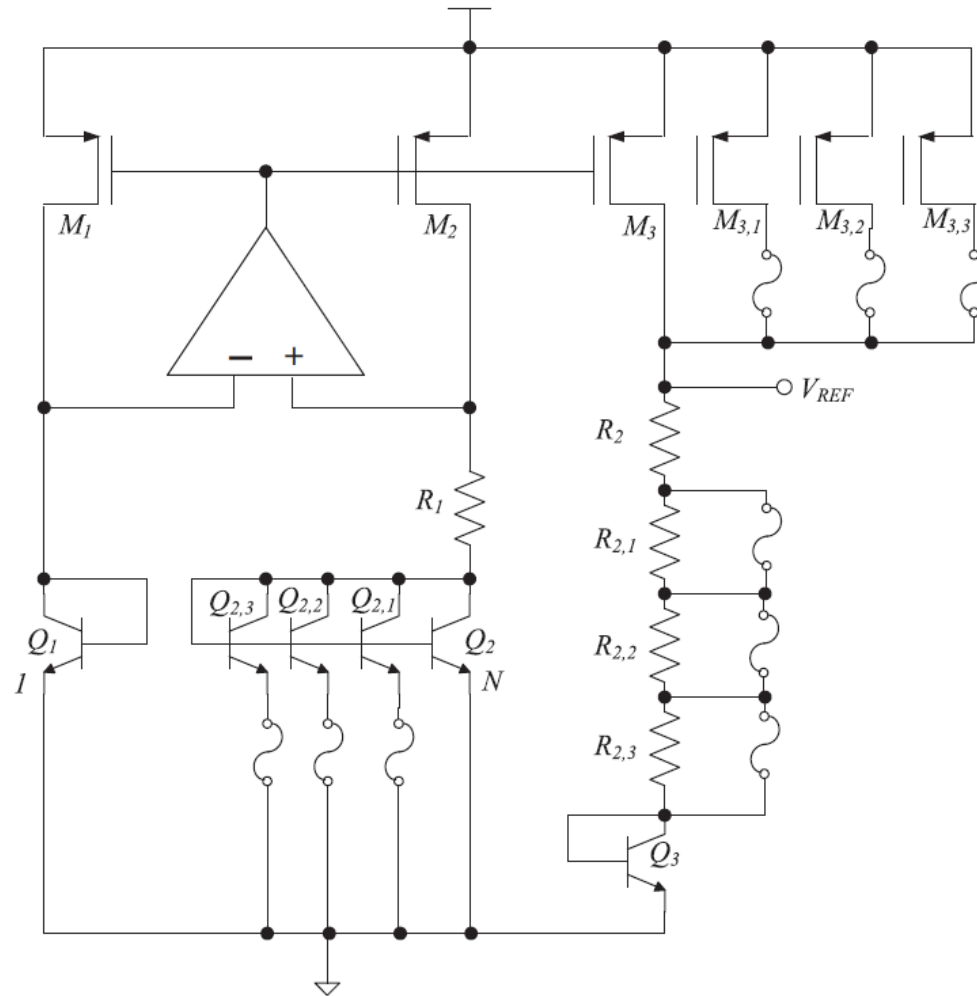


$$\Delta V_{REF,R} = \frac{R_2}{R_1} e_R V_T \ln N$$

Error source : Resistor mismatch, solution

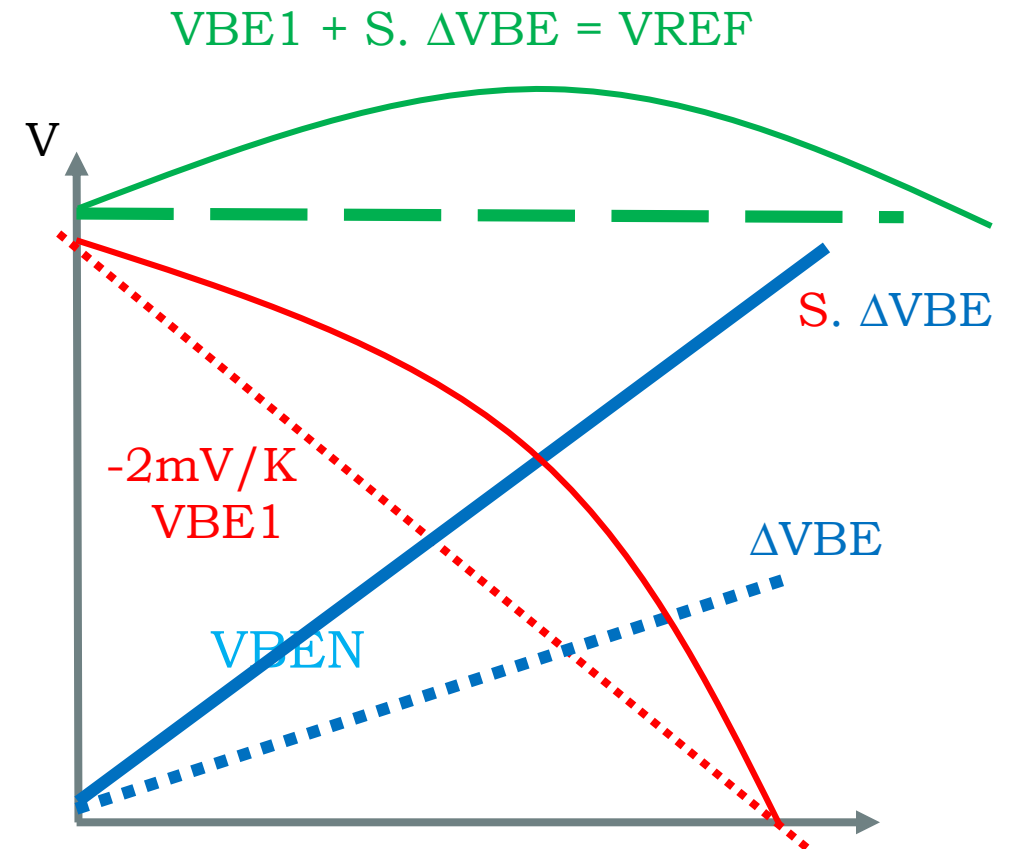
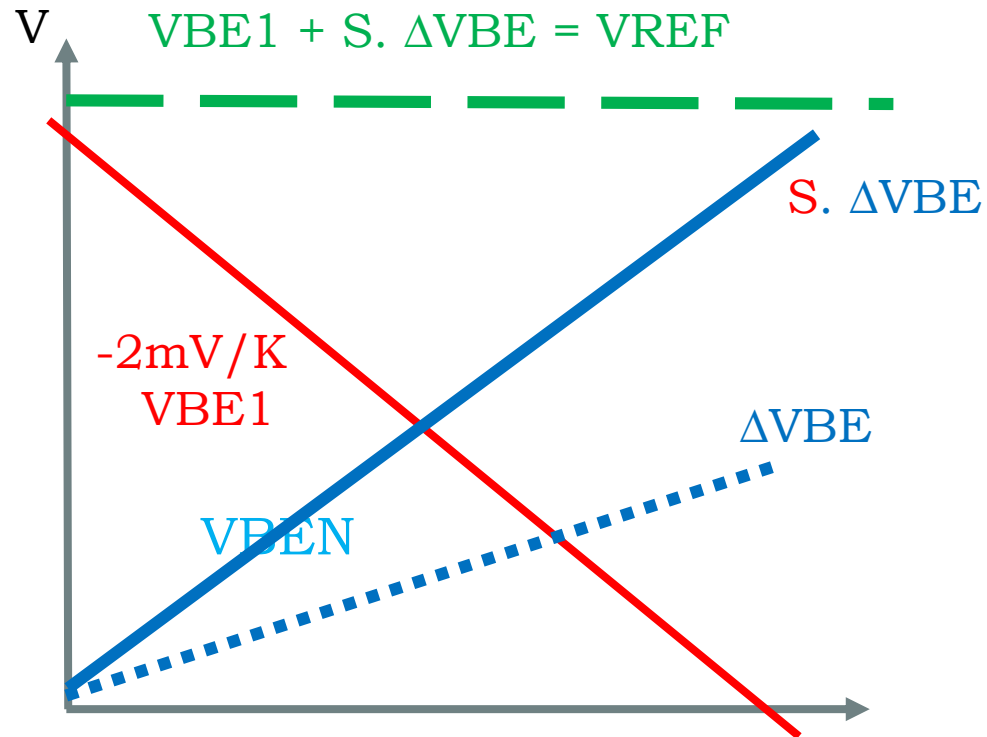
$$\Delta V_{REF,rb} \approx \frac{R_2}{R_1} V_T \ln \left(N \frac{\Delta r_{b1,2}}{\beta R_1} \right) + \frac{R_2}{R_1} V_T \ln \left(N \frac{r_{b3}}{\beta R_2} \right)$$

Trimming for removal of error sources



Curvature Correction of Voltage Reference !!!

Curvature concept



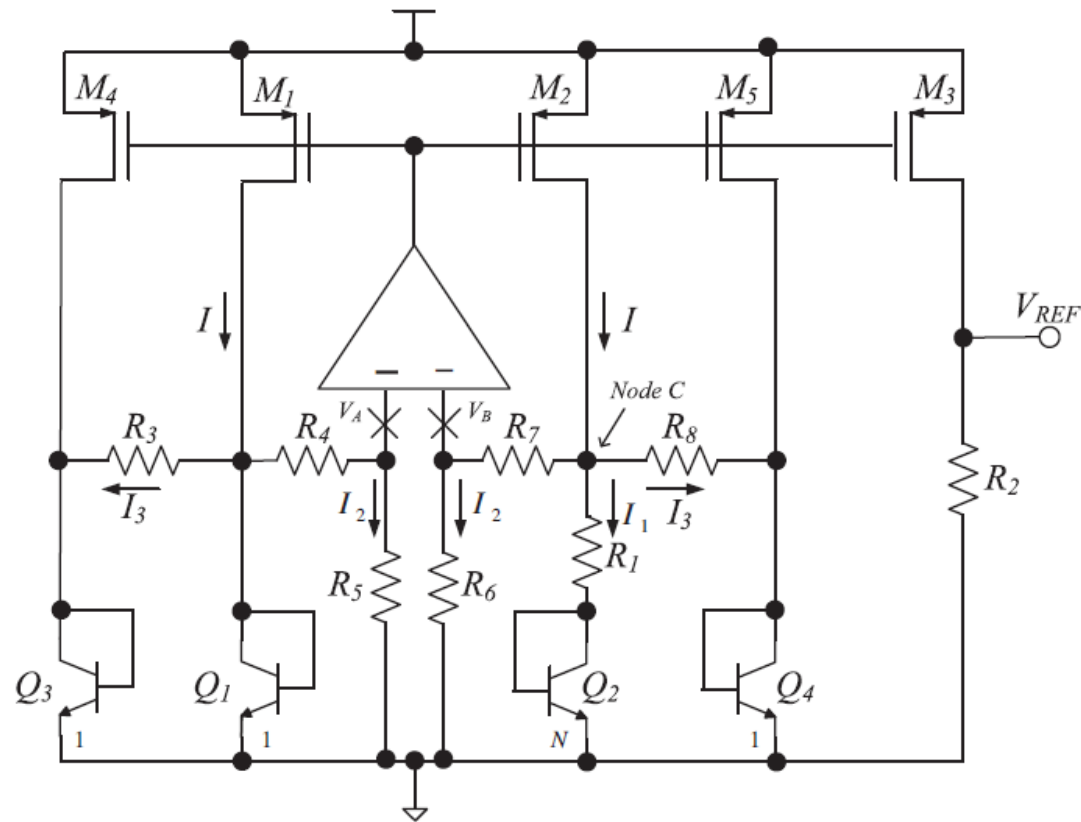
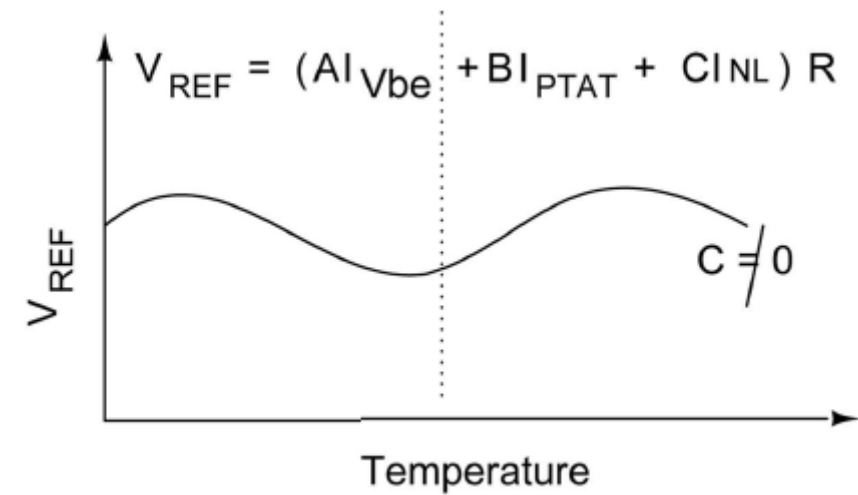
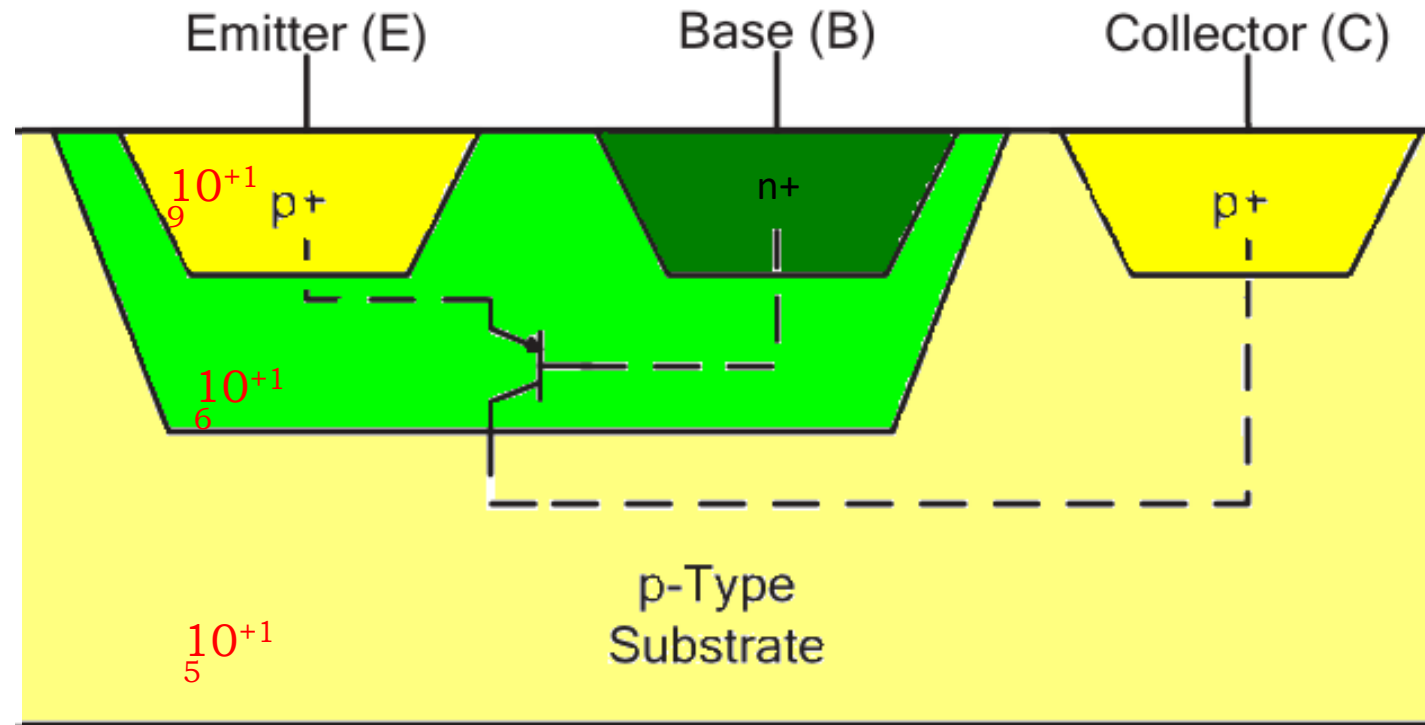


Figure The schematic of opamp based β -multiplier bandgap voltage reference circuit with BJT current subtraction to achieve second order curvature correction (Gunawan *et al.*, 1993).

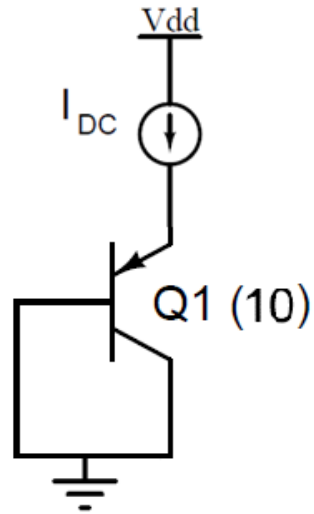


V_{REF} by multiplying β with V_T

Parasitic BJT :

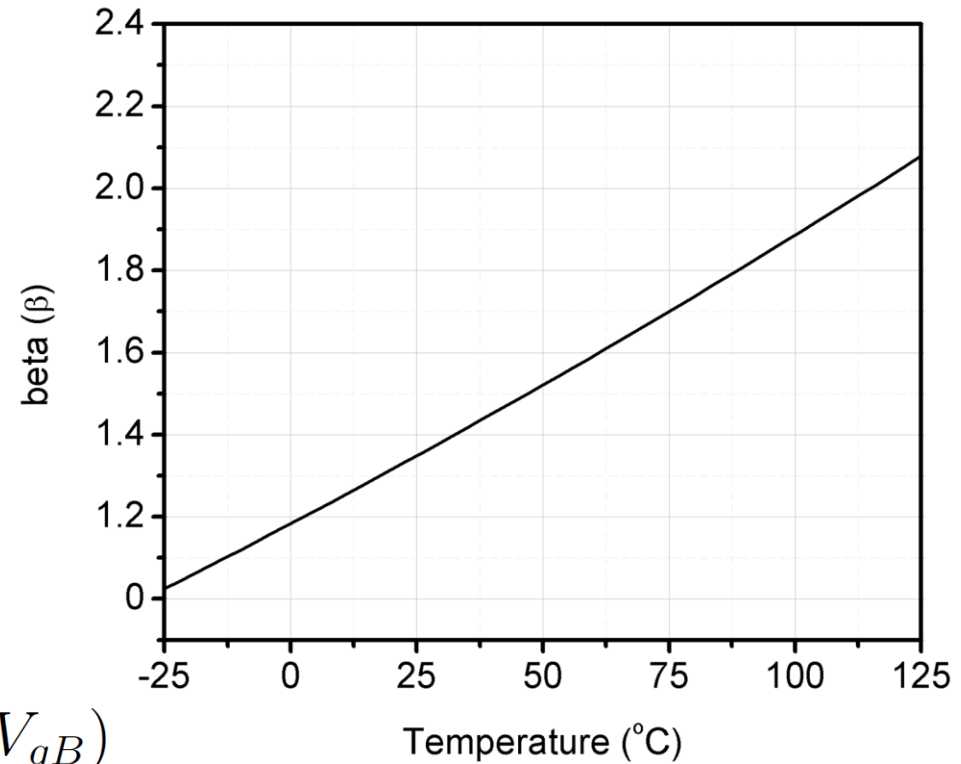


2. V_{REF} by multiplying β with V_T



$$\beta = \frac{D_B n_{iB}^2 N_E L_E}{D_E n_{iE}^2 N_B W_B}$$

$$= \frac{D_B N_E L_E}{D_E N_B W_B} \exp \frac{(V_{gE} - V_{gB})}{V_T}$$



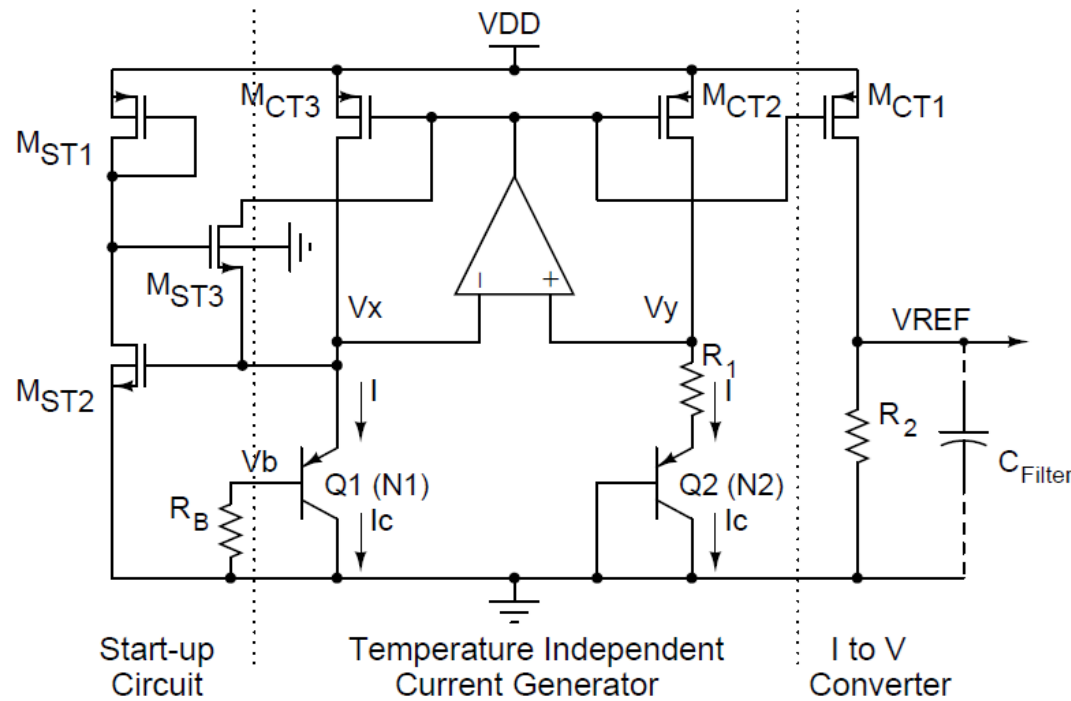
(Tang TED-80, Wang-PhD Des05)

$$V_{gE} < V_{gB} \quad (V_{gE} - V_{gB}) \approx -52 \text{ mV at } 300 \text{ K}$$

Therefore β is exponentially dependant on temperature.



2. V_{REF} by multiplying β with V_T



$$V_{E1} = V_T \ln \frac{I_C}{N_1 I_S} + R_B I_B$$

$$= V_T \ln \frac{I_C}{N_1 I_S} + R_B \frac{I}{(\beta + 1)}$$

$$V_{E2} = V_T \ln \frac{I_C}{N_2 I_S}$$

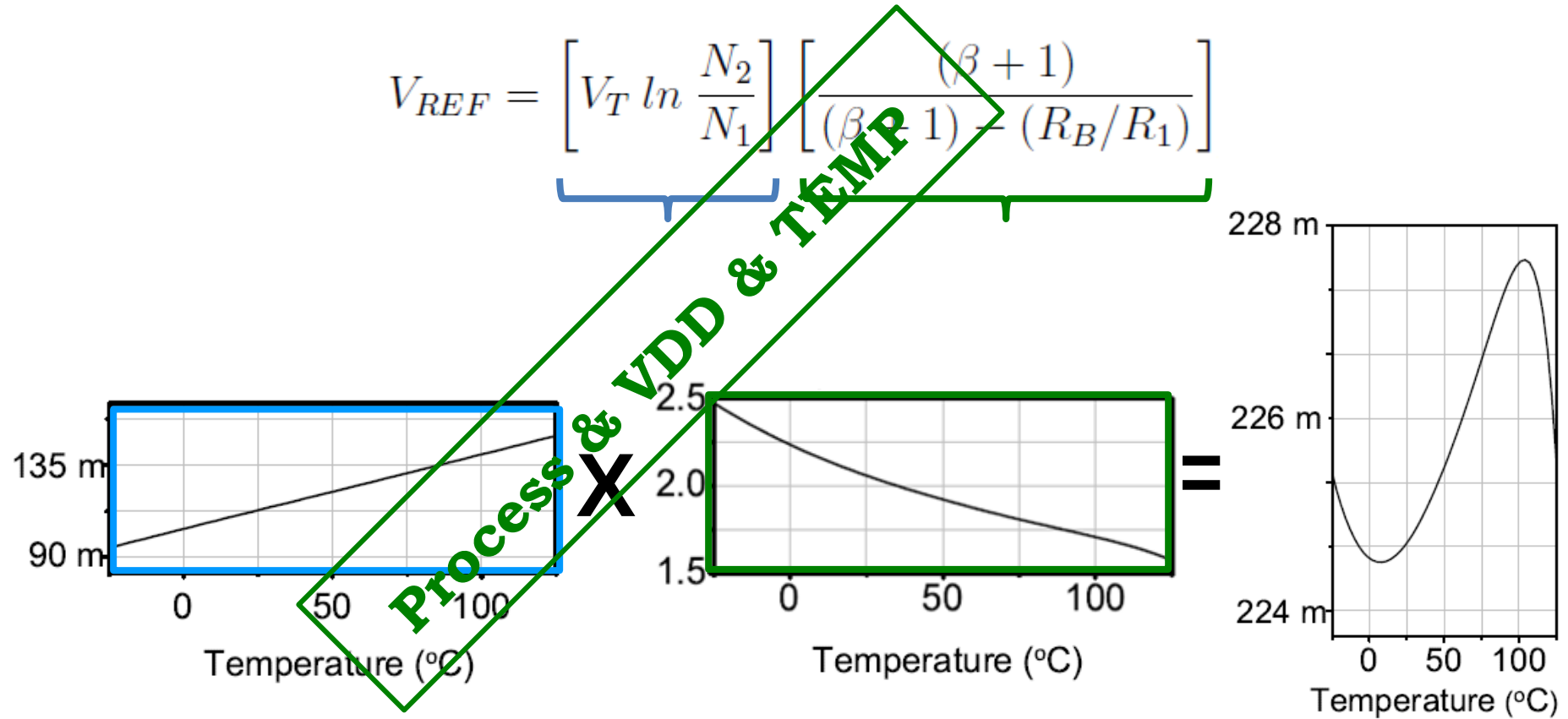
$$\Delta V_E = V_{E1} - V_{E2}$$

$$= V_T \ln \frac{N_2}{N_1} + R_B \frac{I}{(\beta + 1)}$$

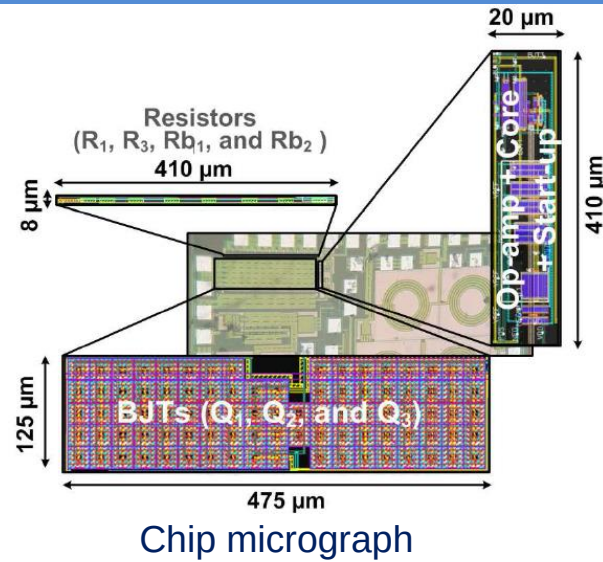
$$R_1 = R_2, \quad V_{REF} = \Delta V_E = \left[V_T \ln \frac{N_2}{N_1} \right] \left[\frac{(\beta + 1)}{(\beta + 1) - (R_B/R_1)} \right]$$



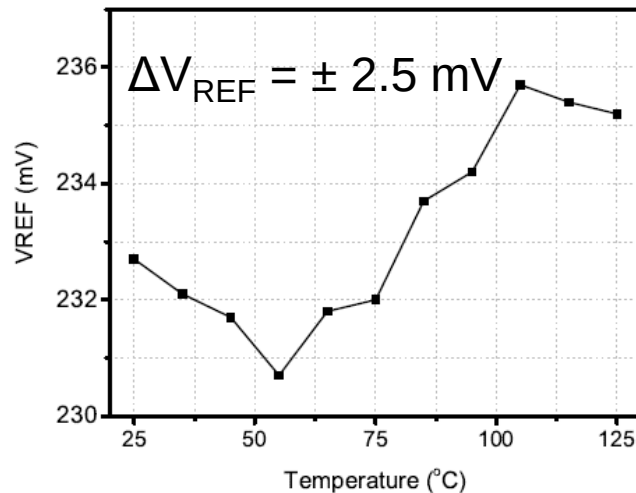
2. V_{REF} by multiplying β with V_T



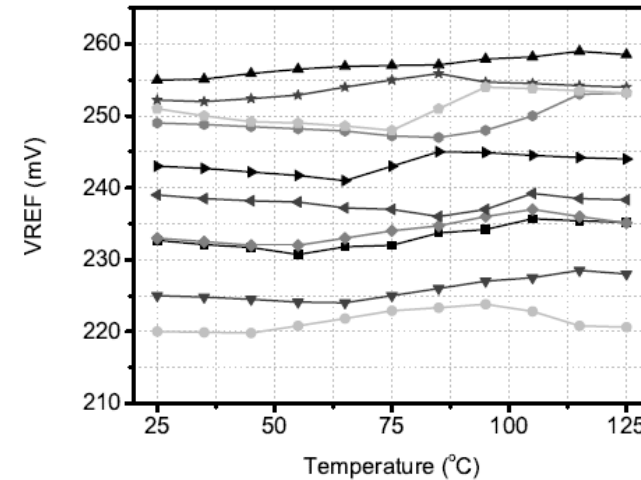
2. V_{REF} by multiplying β with V_T



Area = 0.072 mm²



Reference voltage with temperature



V_{REF} spread = $\pm 19.8 \text{ mV}$ (10 samples)



2. V_{REF} by multiplying β with V_T

Performance comparison of proposed voltage reference circuits

Architecture	$\Delta V_{TH} - V_T$ Core-IO (Measured)	$\Delta V_{TH} - V_T$ Core-Core (Measured)	$\beta - V_T$ (Measured)
Technology	180 nm	180 nm	180 nm
V_{DD} (V)	1	1	0.8
I_{DD} (A)	15μ	1.13μ	3.1μ
Power Cons. (W)	15μ	1.13μ	2.48μ
ΔV_{REF} (V)	± 2.2 m	± 1.225 m	± 2.5 m
MAPV (%)	0.79	1.43	2.13
Temp. Range ($^{\circ}$ C)	25 : 125	25 : 125	25 : 125
Area (mm ²)	0.028	0.0048	0.072
Voltage Spread (V)	± 90 m	± 14 m	± 19.8 m



Ker et al. JSSC`06

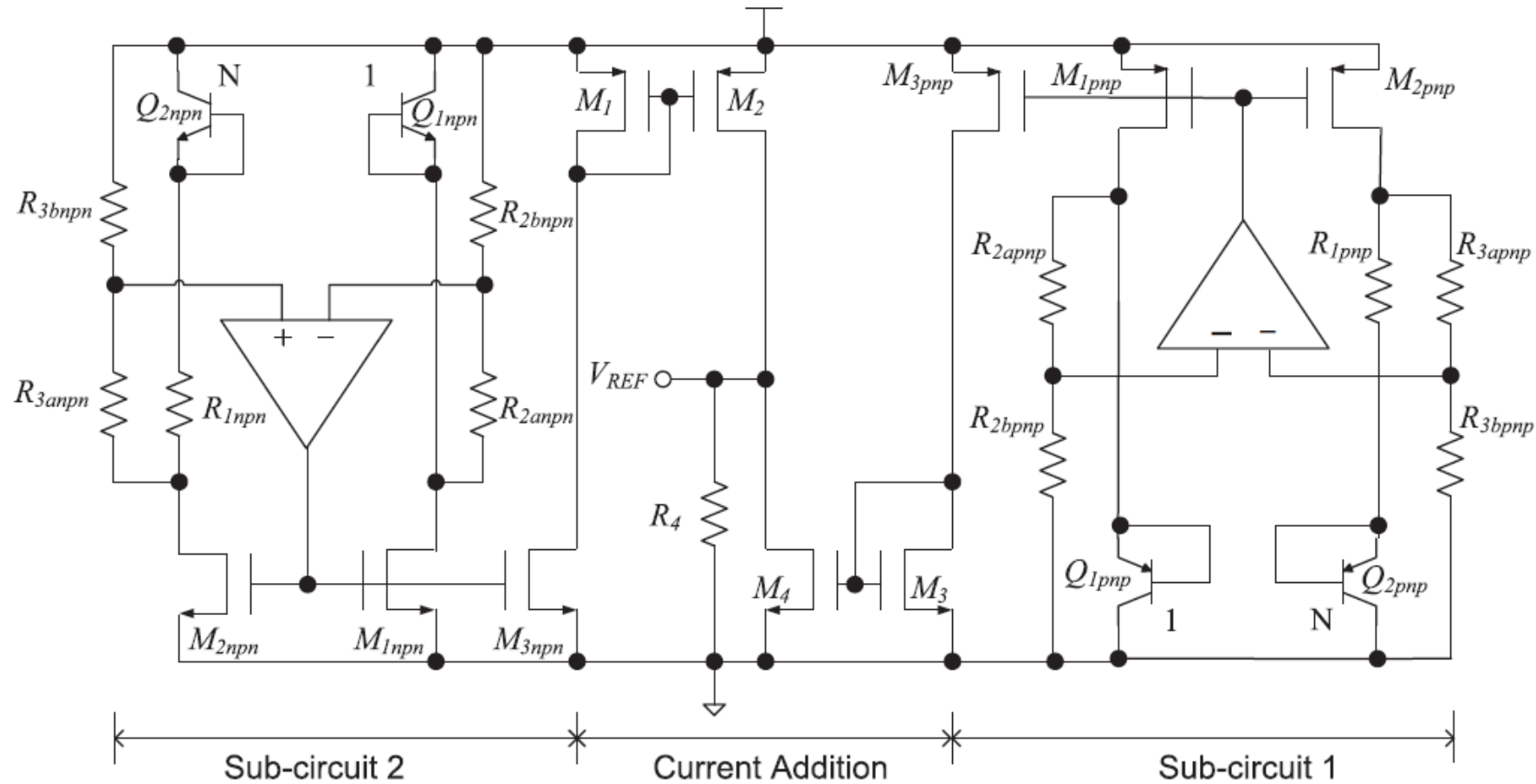


Figure Sub-1V bandgap voltage reference by resistive sub-division with NMOS transistors current mirror based β -multiplier, and the complete schematic (Ker *et al.*, 2006).

Hande et al. JOS`15

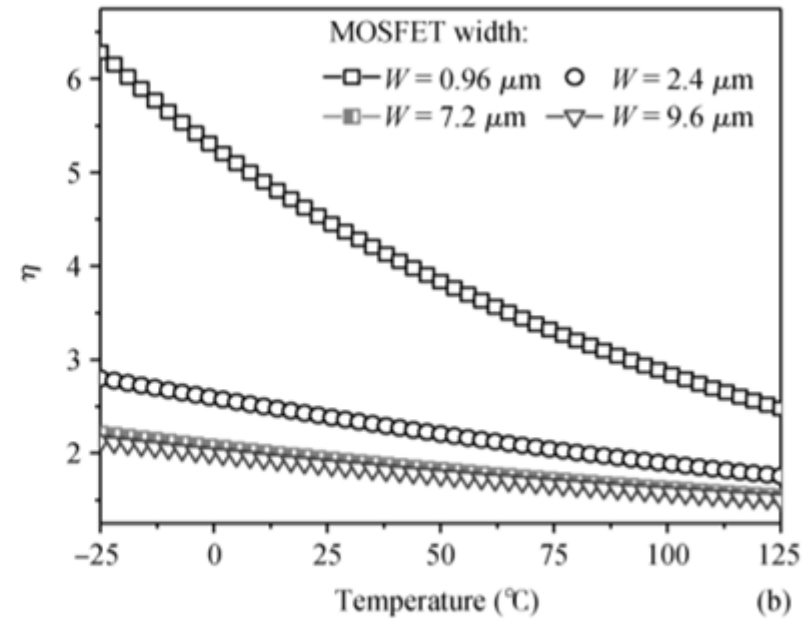
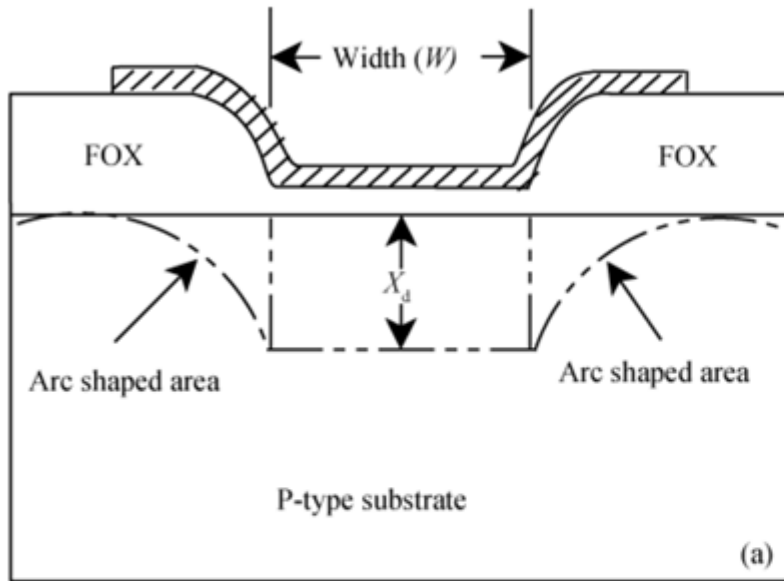
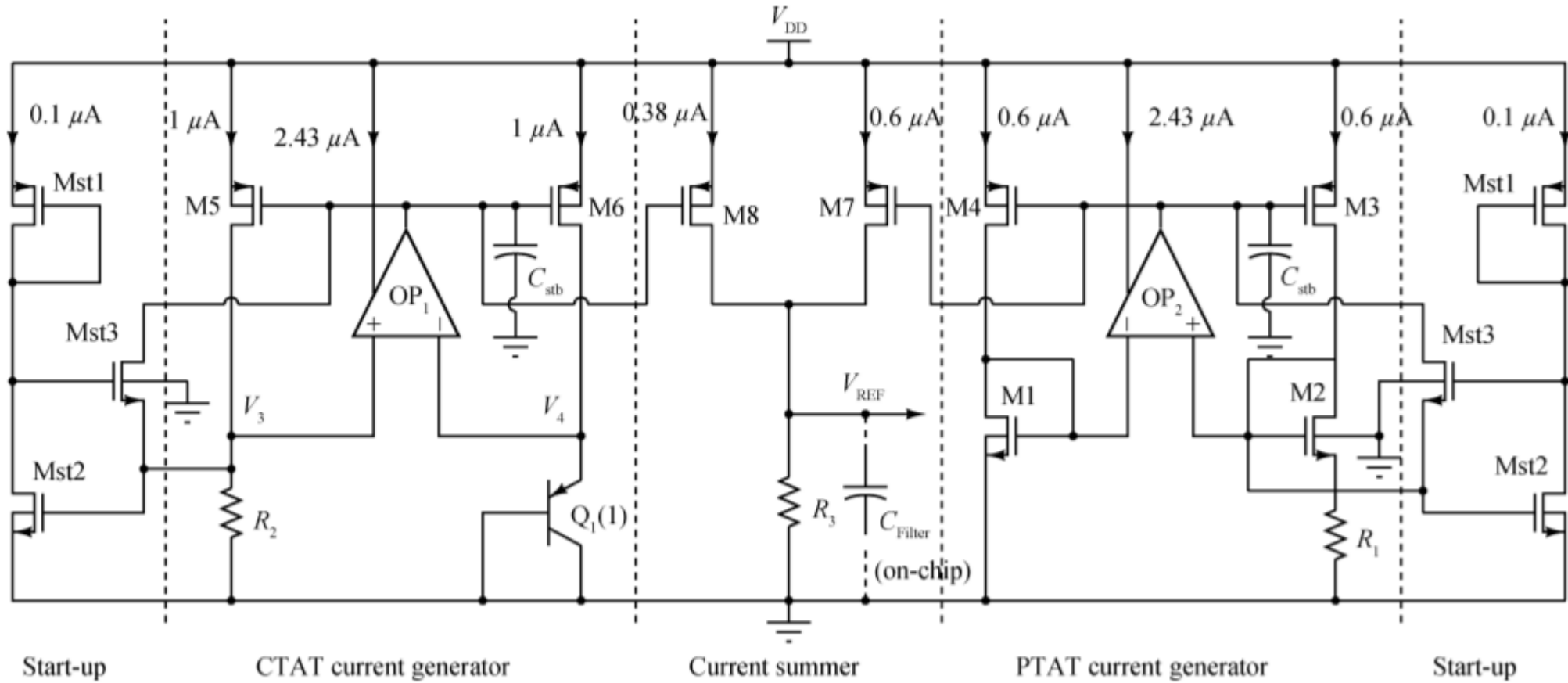


Figure . (a) NMOS transistor structure in the width direction for step transition isolation. (b) Variation in the η with varying temperature for different widths.

Hande et al. JOS'15



Hande et al. JOS`15

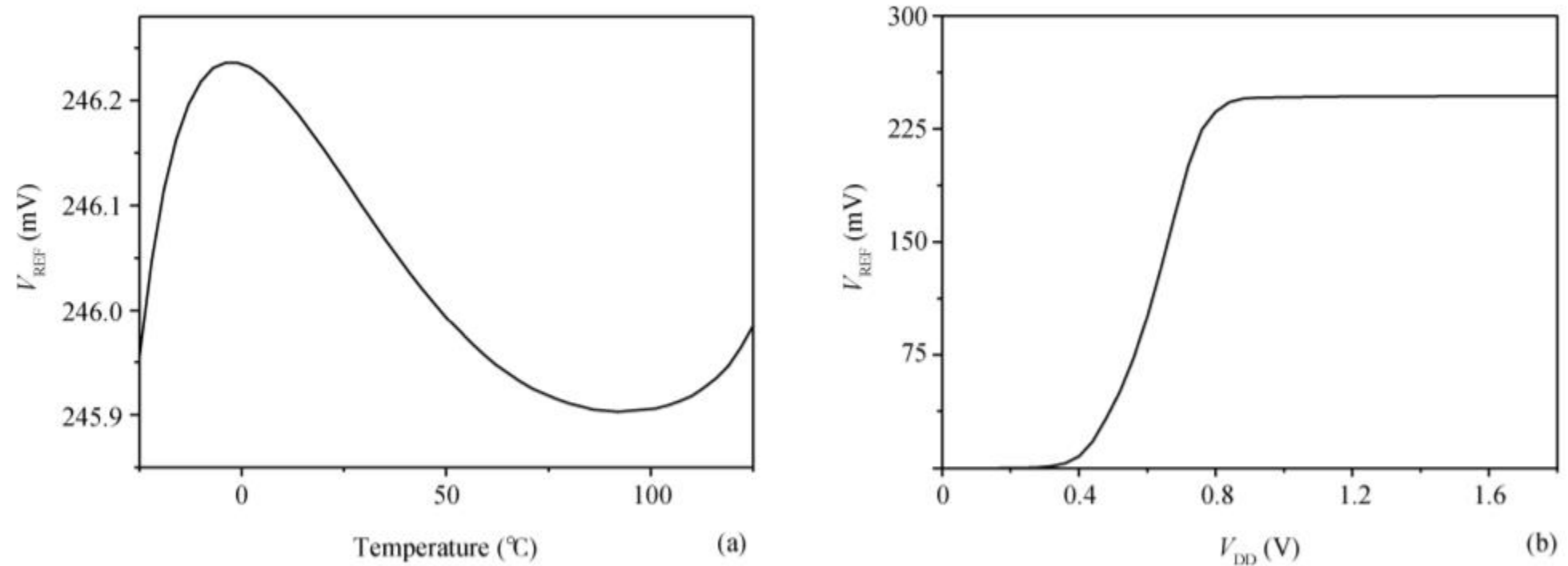
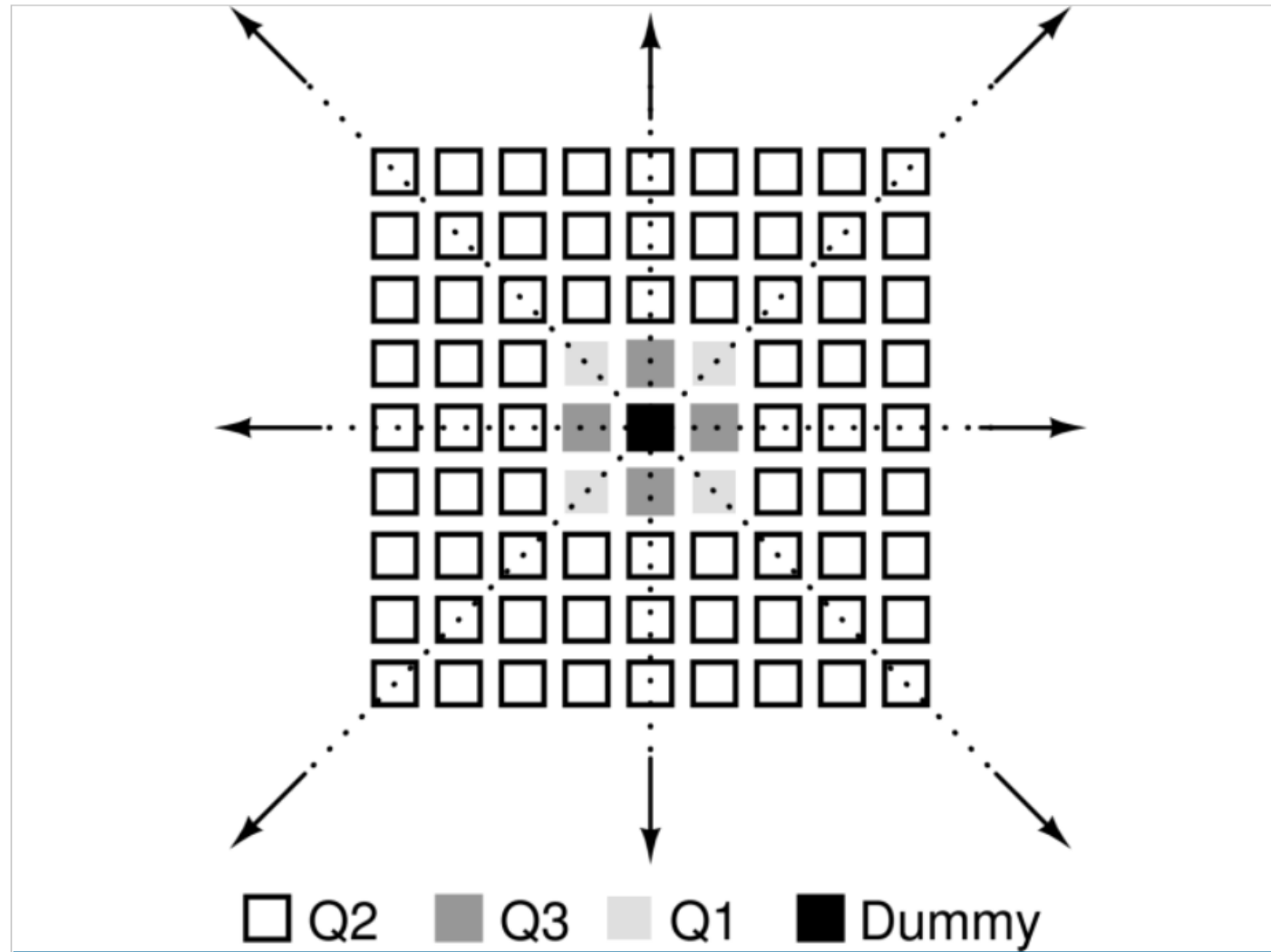


Figure. (a) Reference voltage with varying temperature. (b) Reference voltage with varying supply voltage.

Layout Technique

Symmetry in layout is important



Reference Book

CMOS VOLTAGE REFERENCES AN ANALYTICAL AND PRACTICAL PERSPECTIVE

By Chi-Wah Kok and Wing-Shan Tam

Canaan Microelectronics Corporation Ltd, Hong Kong

Conclusion

Voltage reference design is one of the design skill which tells us about DC to AC analysis in simpler way...

ANALOG IS FUN !!!...